



Acoustics of porous media

Lecture 2: Equivalent fluid models

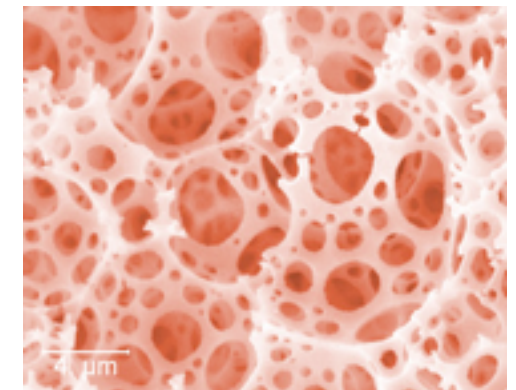
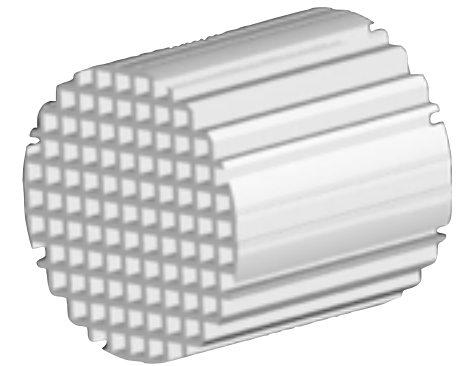
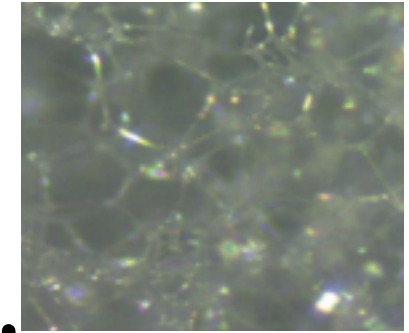
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- **Skeleton is motionless**
 - Nor displacement, nor strain
 - Also called rigid frame
 - Influence through microscopic geometry 
- **Propagation in a dissipative fluid**
 - Viscous effects
 - Thermal effects
- **Three different type of models**
 - Empirical
 - Geometrical
 - Semi phenomenological



Equivalent fluid displacement

$$\mathbf{u}^{eq} = \langle \mathbf{u}_\mu^f \rangle_\Omega = \phi \mathbf{u}^f$$

Fluid dilatation

$$\varepsilon^{eq} = \nabla \cdot \mathbf{u}^{eq}$$

Interstitial pressure

$$p = \langle p_\mu \rangle_{\Omega_f}$$

Isotropic stress

Are these two fields sufficient ?

- ✓ Thermodynamics/ Mechanics decoupling
- ✓ Dissipation effects (Temperature)
- ✓ Compression effects (Density)
- ✓ It would be nice !

Let's have a look at the microscopical physics

6 microscopic equations on fields fluctuation

Kirchhoff theory - 1868
Stinson JASA 89(2) - 1981

- Linearized Navier-Stokes equation:

$$j\omega\rho_0\vec{v} = -\nabla p + \left(\frac{4}{3}\eta + \zeta\right) \nabla \nabla \cdot \vec{v} - \eta \nabla \wedge \nabla \wedge \vec{v}.$$

- Mass conservation equation:

$$j\omega\rho = -\rho_0 \nabla \cdot \vec{v}.$$

- Thermal conduction equation:

$$\kappa \nabla^2 \tau = j\omega \frac{T_0}{P_0} (\rho_0 C_v p - P_0 C_p \rho).$$

- Equation of state for ideal gas

$$p = \frac{P_0}{\rho_0 T_0} (\rho_0 \tau + T_0 \rho).$$

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**Velocity
(vector)**

- Mass conservation equation:

$$j\omega\rho = -\rho_0 \nabla \cdot \vec{v}.$$

**Fluid density
(scalar)**

- Thermal conduction equation:

$$\kappa \nabla^2 \tau = j\omega \frac{T_0}{P_0} (\rho_0 C_v p - P_0 C_p \rho).$$

**Fluid temperature
(scalar)**

- Equation of state for ideal gas

**Fluid pressure
(scalar)**

$$p = \frac{P_0}{\rho_0 T_0} (\rho_0 \tau + T_0 \rho).$$

6 microscopic equations on fields fluctuation

Kirchhoff theory - 1868
Stinson JASA 89(2) - 1981

- Linearized Navier-Stokes equation:

- ρ_0 : Air density ($= 1.17 \text{ kg m}^{-3}$)
 - T_0 : Ambient temperature ($= 25^\circ\text{C}$)
 - P_0 : Ambient pressure ($= 1013523 \text{ Pa}$)
 - η : Shear viscosity ($= 1.84 \cdot 10^{-5} \text{ kg m}^{-1} \text{ s}^{-1}$)
 - ζ : Volume viscosity (*neglected*)
 - κ : Thermal conductivity ($= 2.6 \cdot 10^{-2} \text{ W m}^{-1} \text{ K}^{-1}$)
 - C_p : Specific heat at constant pressure ($= 1.006 \cdot 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$)
 - C_v : Specific heat at constant volume ($= 7.18 \cdot 10^2 \text{ J kg}^{-1} \text{ K}^{-1}$)
- (scalar)
- $p = \frac{\rho_0 T_0}{\rho_0 T_0} (\rho_0 T_0 + \rho_0 p)$

6 microscopic equations on fields fluctuation

Kirchhoff theory - 1868
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- Linearized Navier-Stokes equation:

$$j\omega\rho_0\vec{v} = -\nabla p + \left(\frac{4}{3}\eta + \zeta\right) \nabla \nabla \cdot \vec{v} - \eta \nabla \wedge \nabla \wedge \vec{v}.$$

- Mass conservation equation:

**Elimination of density
and pressure**

$$j\omega\rho = -\rho_0 \nabla \cdot \vec{v}.$$

**Velocity
(vector)**

**Fluid density
(scalar)**

- Thermal conduction equation:

$$\kappa \nabla^2 \tau = j\omega \frac{T_0}{P_0} (\rho_0 C_v p - P_0 C_p \rho).$$

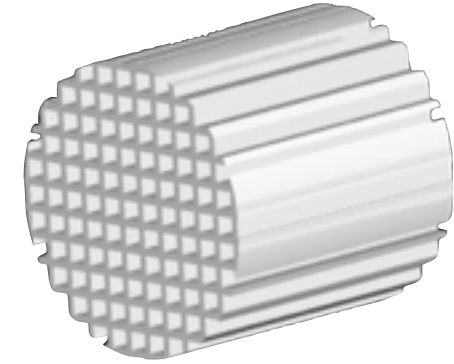
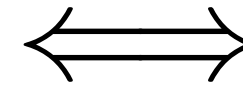
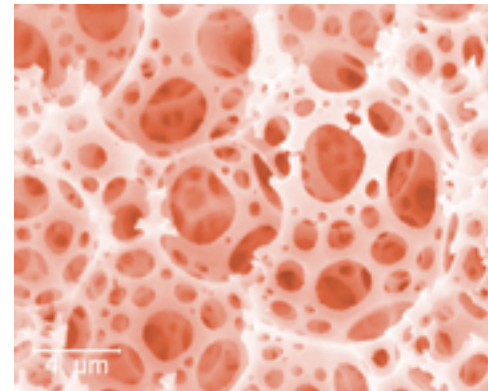
**Fluid temperature
(scalar)**

- Equation of state for ideal gas

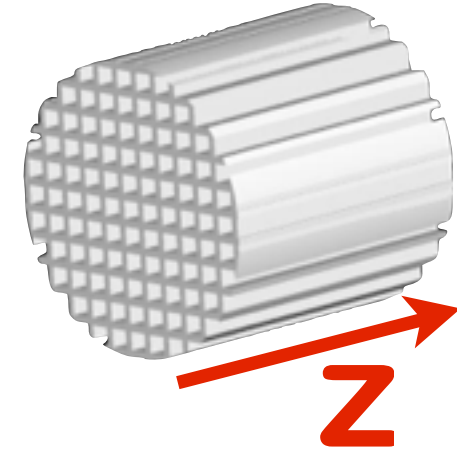
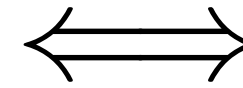
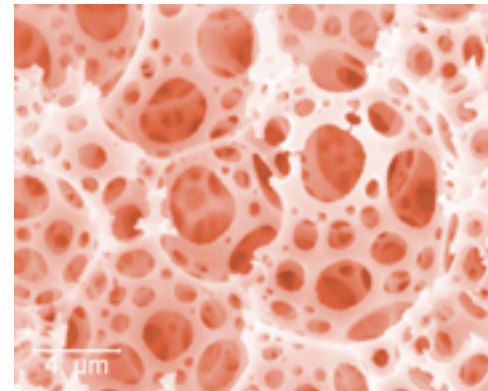
**Fluid pressure
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$$p = \frac{P_0}{\rho_0 T_0} (\rho_0 \tau + T_0 \rho).$$

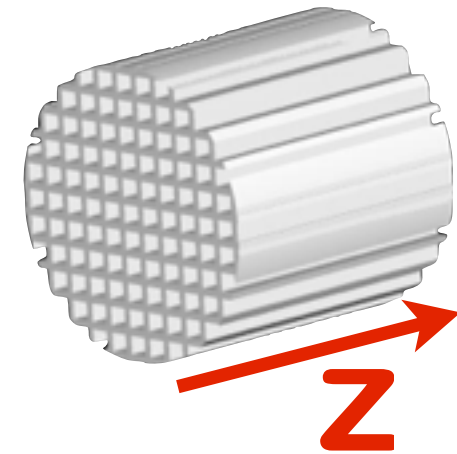
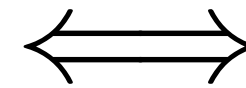
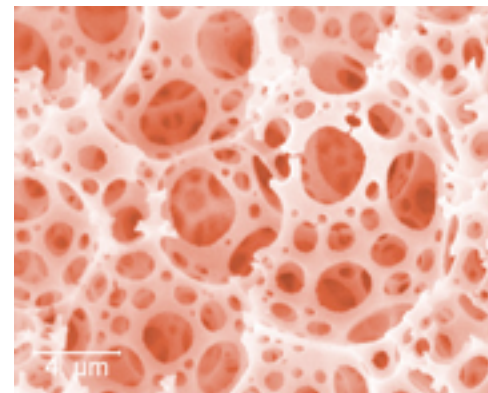
Propagation in a waveguide



Propagation in a waveguide



Propagation in a waveguide



Separation of variables

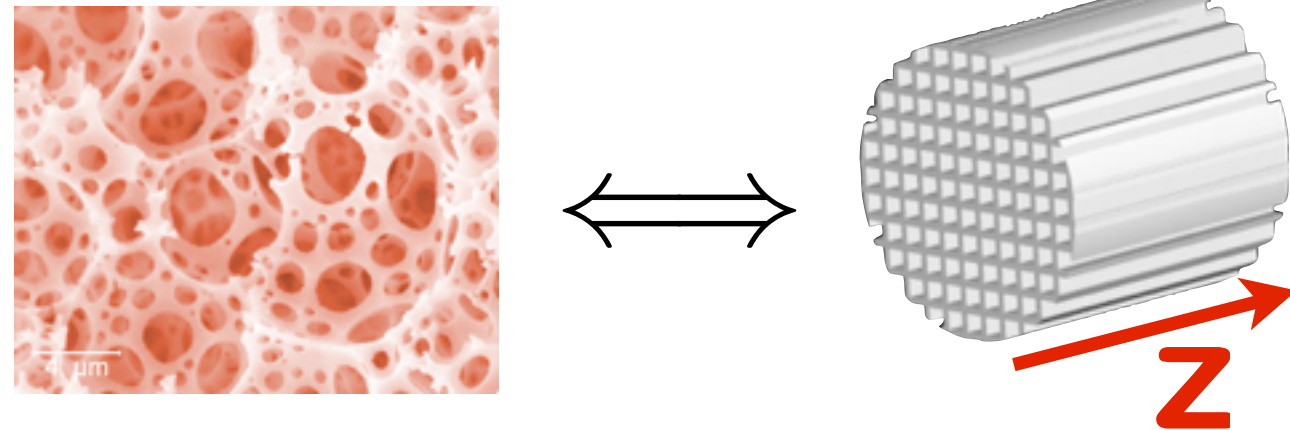
$$X(x, y, z) = X(x_S) e^{i\delta z}$$

Section of the tube

Wave number
(unknown)

$$\delta = ?$$

Propagation in a waveguide



Separation of variables

$$X(x, y, z) = X(x_S) e^{i\delta z}$$

Section of the tube

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Boundary conditions

- ✓ Motionless skeleton : $\mathbf{v}_\Gamma = 0$
- ✓ Isothermal skeleton : $\tau_\Gamma = 0$

Navier Stokes:
$$j\omega\vec{v} = -\frac{c_0^2}{\gamma T_0} \nabla \tau - \nu \nabla \wedge \nabla \wedge \vec{v}$$

Thermal equation:
$$\nu' \nabla^2 \tau = j\omega \tau + (\gamma - 1) T_0 \nabla \cdot \vec{v}$$

$$\nu' = \frac{\kappa}{\rho_0 C_v}$$
$$\zeta = \frac{4}{3}\nu + \frac{c_0^2}{j\omega\gamma}$$

Navier Stokes: $\nabla \cdot j\omega \vec{v} = -\frac{c_0^2}{\gamma T_0} \nabla \tau - \nu \nabla \wedge \nabla \wedge \vec{v}$

Thermal equation: $\nu' \nabla^2 \tau = j\omega \tau + (\gamma - 1) T_0 \nabla \cdot \vec{v}$

$$\nu' = \frac{\kappa}{\rho_0 C_v}$$

$$\zeta = \frac{4}{3} \nu + \frac{c_0^2}{j\omega \gamma}$$

$$\left(\zeta' \nu' \nabla^2 \nabla^2 - \left(c_0^2 + j\omega \left[\frac{4}{3} \nu + \nu' \right] \right) \nabla^2 - \omega^2 \right) \tau = 0$$

Navier Stokes: $\nabla \cdot j\omega \vec{v} = -\frac{c_0^2}{\gamma T_0} \nabla \tau - \nu \nabla \wedge \nabla \wedge \vec{v}$

Thermal equation: $\nu' \nabla^2 \tau = j\omega \tau + (\gamma - 1) T_0 \nabla \cdot \vec{v}$

$$\nu' = \frac{\kappa}{\rho_0 C_v}$$

$$\zeta = \frac{4}{3} \nu + \frac{c_0^2}{j\omega \gamma}$$

$$\zeta \nu' \lambda^2 - \left(c_0^2 + j\omega \left[\frac{4}{3} \nu + \nu' \right] \right) \lambda - \omega^2 = 0$$

$$(\nabla^2 - \lambda_1)(\nabla^2 - \lambda_2)\tau = 0 \Rightarrow \tau = \tau_1 + \tau_2$$

$$(\nabla^2 - \lambda_1)\tau_1 = 0, \quad (\nabla^2 - \lambda_2)\tau_2 = 0$$

Navier Stokes: $\nabla \cdot j\omega \vec{v} = -\frac{c_0^2}{\gamma T_0} \nabla \tau - \nu \nabla \wedge \nabla \wedge \vec{v}$

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$$\nu' = \frac{\kappa}{\rho_0 C_v}$$

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$$(\nabla^2 - \lambda_1)\tau_1 = 0, \quad (\nabla^2 - \lambda_2)\tau_2 = 0$$

$$j\omega \vec{v} - \zeta \nabla \nabla \cdot \vec{v} + \nu_0 \nabla \wedge \nabla \wedge \vec{v} = -\frac{c_0^2}{\gamma T_0} \nabla \tau$$

$$\vec{v} = \vec{v}_g + \underbrace{\vec{v}_1 + \vec{v}_2}_{\vec{v}_p}$$

$$\tau(r, \theta, z) = \tau(r, \cancel{\theta})e^{\delta z} = \tau(r)e^{\delta z}$$

Invariance by rotation



$$\tau(r, \theta, z) = \tau(r, \cancel{\theta})e^{\delta z} = \tau(r)e^{\delta z}$$

Invariance by rotation

Thermal equation:

$$\tau_i''(r) + \frac{1}{r}\tau_i'(r) + (\delta^2 - \lambda_i)\tau_i(r) = 0, \quad i = 1, 2$$

$$\tau(r, \theta, z) = \tau(r, \cancel{\theta})e^{\delta z} = \tau(r)e^{\delta z}$$

Invariance by rotation

Thermal equation:

$$\tau_i''(r) + \frac{1}{r}\tau_i'(r) + (\delta^2 - \lambda_i)\tau_i(r) = 0, \quad i = 1, 2$$

Definition : $W_\alpha(r) = J_0(r\sqrt{\delta^2 - \alpha})$

Solution of Bessel equation

$$W_\alpha'' + \frac{W_\alpha'}{r} + (\delta^2 - \alpha)W_\alpha = 0$$

General form for temperature:

$$\tau(r) = c_1 W_{\lambda_1}(r) + c_2 W_{\lambda_2}(r)$$

$c_1 = ?$

$c_2 = ?$

$$v_r(r, \theta, z) = v_r(r)e^{\delta z}, \quad v_z(r, \theta, z) = v_z(r)e^{\delta z}$$

Radial and longitudinal Navier-Stokes equations:

$$\begin{aligned} j\omega v_r &= \boxed{-\frac{c_0^2}{\gamma T_0} \tau'} + \zeta \left(v_r'' + \frac{v_r'}{r} - \frac{v_r}{r^2} + \delta v_z' \right) + \delta \nu (\delta v_r - v_z'), \\ j\omega v_z &= \boxed{-\frac{c_0^2 \delta}{\gamma T_0} \tau} + \zeta \left(\delta \frac{v_r}{r} + \delta v_r' + m^2 v_z \right) - \nu (\delta v_r' - v_z'' + \delta \frac{v_r}{r} - \frac{v_z'}{r}) \end{aligned}$$

Source **Coupled Bessel equations**

$$v_r(r, \theta, z) = v_r(r)e^{\delta z}, \quad v_z(r, \theta, z) = v_z(r)e^{\delta z}$$

Radial and longitudinal Navier-Stokes equations:

$$j\omega v_r = \boxed{-\frac{c_0^2}{\gamma T_0} \tau'} + \zeta \left(v_r'' + \frac{v_r'}{r} - \frac{v_r}{r^2} + \delta v_z' \right) + \delta \nu (\delta v_r - v_z'),$$
$$j\omega v_z = \boxed{-\frac{c_0^2 \delta}{\gamma T_0} \tau} + \zeta \left(\delta \frac{v_r}{r} + \delta v_r' + m^2 v_z \right) - \nu (\delta v_r' - v_z'' + \delta \frac{v_r}{r} - \frac{v_z'}{r})$$

Source

Coupled Bessel equations

General form of solutions:

$$v_r(r) = a_v W_{\lambda_v}'(r) + a_1 W_{\lambda_1}'(r) + a_2 W_{\lambda_2}'(r),$$

$$v_z(r) = b_v W_{\lambda_v}(r) + b_1 W_{\lambda_1}(r) + b_2 W_{\lambda_2}(r)$$

$$\lambda_v = ?, \quad a_v = ?, \quad b_v = ?, \quad a_i = ?, \quad b_i = ?$$

$$j\omega = \zeta \left(\lambda_v - \delta^2 + \delta \frac{b_v}{a_v} \right) + \delta \nu \left(\delta - \frac{b_v}{a_v} \right),$$

$$j\omega \frac{b_v}{a_v} = \zeta \delta \left(\lambda_v - \delta^2 + \delta \frac{b_v}{a_v} \right) - \nu (\lambda_v - \delta^2) \left(\delta - \frac{b_v}{a_v} \right)$$

Two coupled non-linear equations (with trivial solutions !)

$$\lambda_v = \frac{j\omega}{\nu}, \quad \frac{b_v}{a_v} = \delta - \frac{j\omega}{\delta \nu}$$

$$j\omega = \zeta \left(\lambda_v - \delta^2 + \delta \frac{b_v}{a_v} \right) + \delta \nu \left(\delta - \frac{b_v}{a_v} \right),$$

$$j\omega \frac{b_v}{a_v} = \zeta \delta \left(\lambda_v - \delta^2 + \delta \frac{b_v}{a_v} \right) - \nu (\lambda_v - \delta^2) \left(\delta - \frac{b_v}{a_v} \right)$$

Two coupled non-linear equations (with trivial solutions !)

$$\lambda_v = \frac{j\omega}{\nu}, \quad \frac{b_v}{a_v} = \delta - \frac{j\omega}{\delta \nu}$$

Similarly:

$$\frac{b_i}{a_i} = \delta, \quad \frac{c_i}{a_i} = \frac{\gamma T_0 (\zeta \lambda_i - j\omega)}{c_0^2}$$

$$c_1 = ?, \quad c_2 = ?, \quad b_v = ?, \quad \delta = ?$$

$$\begin{Bmatrix} v_r(r) \\ v_z(r) \\ \tau(r) \end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{W'_{\lambda_1}(r)}{\zeta\lambda_1 - j\omega} & \frac{W'_{\lambda_2}(r)}{j\zeta\lambda_2 - \omega} & \frac{\delta W'_{\lambda_v}(r)}{\delta^2 - \lambda_v} \\ \frac{\delta W_{\lambda_1}(r)}{\zeta\lambda_1 - j\omega} & \frac{\delta W_{\lambda_2}(r)}{j\zeta\lambda_2 - \omega} & b_v W_{\lambda_v}(r) \\ \frac{\gamma T_0 W_{\lambda_1}(r)}{c_0^2} & \frac{\gamma T_0 W_{\lambda_2}(r)}{c_0^2} & 0 \end{bmatrix}}_{[\mathbf{M}(\delta, r)]} \begin{Bmatrix} c_1 \\ c_2 \\ b_v \end{Bmatrix}$$

$$\begin{aligned}
 \begin{Bmatrix} v_r(r) \\ v_z(r) \\ \tau(r) \end{Bmatrix} &= \underbrace{\begin{bmatrix} \frac{W'_{\lambda_1}(r)}{\frac{\zeta\lambda_1 - j\omega}{\delta W_{\lambda_1}(r)}} & \frac{W'_{\lambda_2}(r)}{\frac{j\zeta\lambda_2 - \omega}{\delta W_{\lambda_2}(r)}} & \frac{\delta W'_{\lambda_v}(r)}{\delta^2 - \lambda_v} \\ \frac{\zeta\lambda_1 - j\omega}{\gamma T_0 W_{\lambda_1}(r)} & \frac{\zeta\lambda_2 - j\omega}{\gamma T_0 W_{\lambda_2}(r)} & b_v W_{\lambda_v}(r) \\ \frac{c_0^2}{c_0^2} & \frac{c_0^2}{c_0^2} & 0 \end{bmatrix}}_{[\mathbf{M}(\delta, r)]} \begin{Bmatrix} c_1 \\ c_2 \\ b_v \end{Bmatrix} \\
 \downarrow r = R & \\
 \begin{Bmatrix} v_r(R) = 0 \\ v_z(R) = 0 \\ \tau(R) = 0 \end{Bmatrix} &
 \end{aligned}$$

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 \begin{Bmatrix} v_r(R) = 0 \\ v_z(R) = 0 \\ \tau(R) = 0 \end{Bmatrix} & \quad \delta? \Rightarrow ||[\mathbf{M}(\delta, R)]|| = 0 \\
 c_2 = -\frac{W_{\lambda_1}(R)}{W_{\lambda_2}(R)} c_1 \quad c_1 = \frac{(j\omega - \zeta\lambda_1)(j\omega - \zeta\lambda_2)W_{\lambda_v}(R)}{\delta\zeta W_{\lambda_1}(R)(\lambda_2 - \lambda_1)} b_v
 \end{aligned}$$

Only one unknown (OK as linear problem)

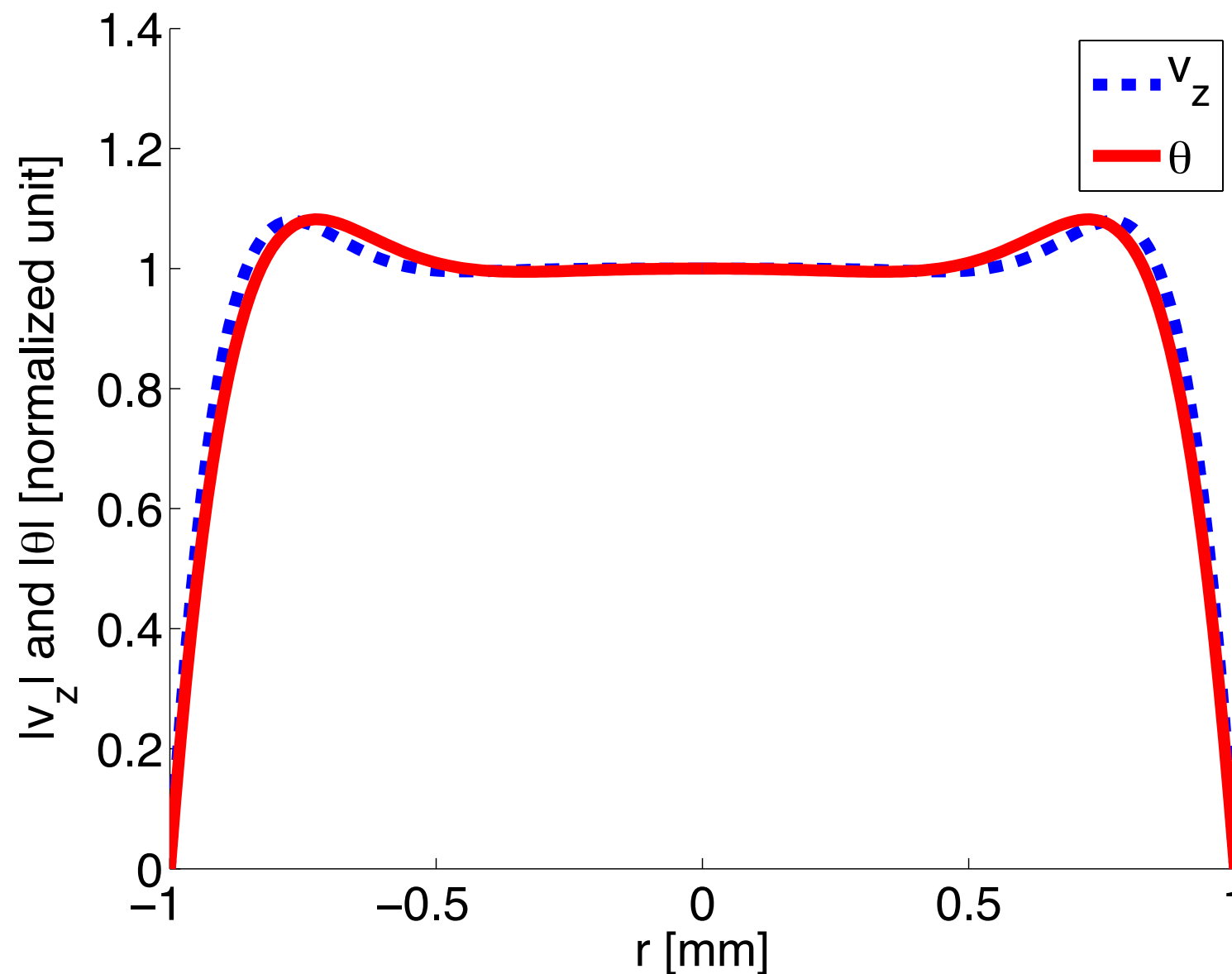
$$v_r, \quad v_z, \quad \tau \Rightarrow p, \quad \rho$$

✓ 500 Hz

✓ Tube of 1mm radius

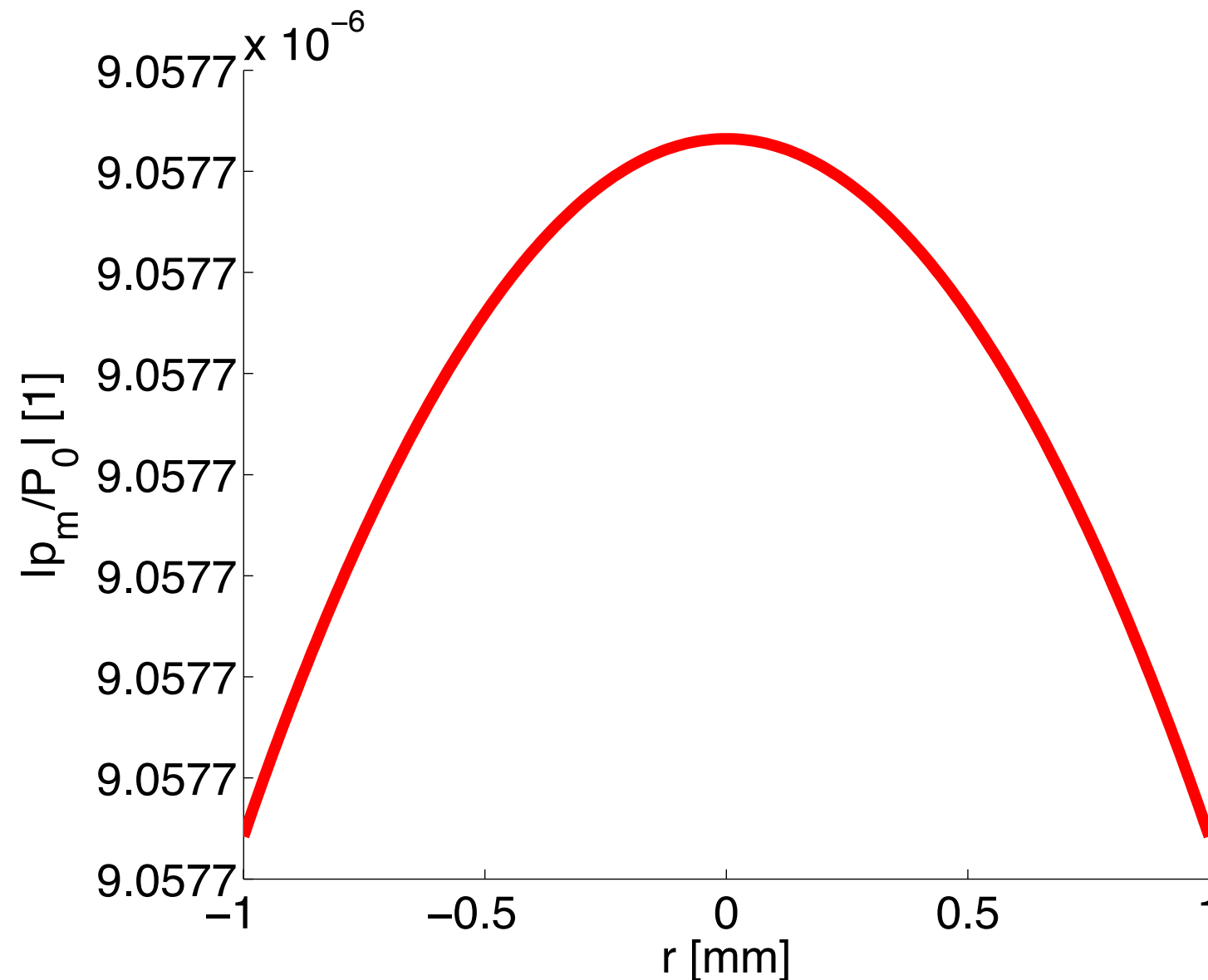
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Pressure constant in the section

- Linearized Navier-Stokes equation:

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$$p(x_S) = C^{te} = \langle p(x_S) \rangle_S$$

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- Equation of state for ideal gas

$$p = \frac{P_0}{\rho_0 T_0} (\rho_0 \tau + T_0 \rho).$$

Two different equations in which pressure act as a force

Navier-Stokes $\nabla \cdot \vec{v} = 0$

Thermal equation $\nabla_{\perp} p = 0$

$$j\omega\rho_0\vec{v} + \eta \nabla \wedge \nabla \wedge \vec{v} = -\nabla p$$

$$\nabla^2 \tau - \frac{j\omega\gamma}{\nu'} \tau = -\frac{j\omega}{\kappa} p$$

$$\nu' = \frac{\kappa}{\rho_0 C_v}$$

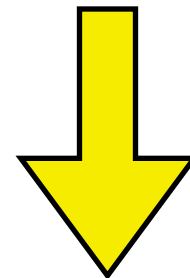
Decoupling of viscous and thermal effects

Two different equations in which pressure act as a force

Navier-Stokes $\nabla \cdot \vec{v} = 0$ $j\omega\rho_0\vec{v} + \eta \nabla \wedge \nabla \wedge \vec{v} = -\nabla p$	Thermal equation $\nabla_{\perp} p = 0$ $\nabla^2 \tau - \frac{j\omega\gamma}{\nu'} \tau = -\frac{j\omega}{\kappa} p$
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$$\nu' = \frac{\kappa}{\rho_0 C_v}$$

Decoupling of viscous and thermal effects



Averaging

$$-\omega^2 \tilde{\rho}_{eq} \mathbf{u}^{eq} = -\nabla p$$

✓ Real part: Inertia
 ✓ Imaginary part: viscous loss

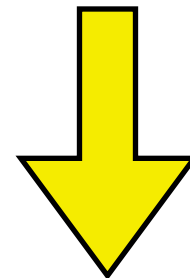
$$p = -\tilde{K}_{eq} \nabla \cdot \mathbf{u}^{eq}$$

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$$(\Delta + \delta_{eq}^2) p = 0$$

Propagation equation of EF (Helmholtz)

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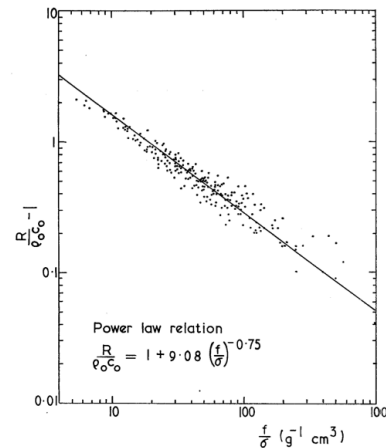
$$\delta_{eq} = \omega \sqrt{\frac{\tilde{\rho}_{eq}}{\tilde{K}_{eq}}} \quad \textbf{Wave number}$$

$$c_{eq} = \sqrt{\frac{\tilde{K}_{eq}}{\tilde{\rho}_{eq}}} \quad \textbf{Celerity}$$

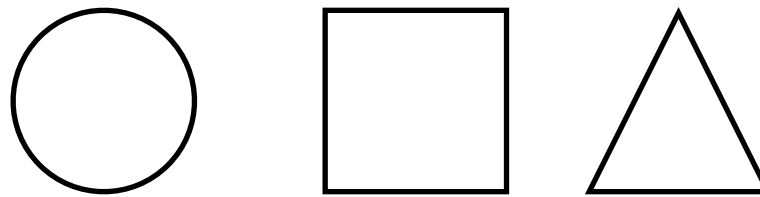
$$Z_{eq} = \rho_{eq} c_{eq} = \sqrt{\tilde{K}_{eq} \tilde{\rho}_{eq}} \quad \textbf{Equivalent fluid impedance}$$

How to find values for $\tilde{\rho}_{eq}$ and $\tilde{K}_{eq}$?

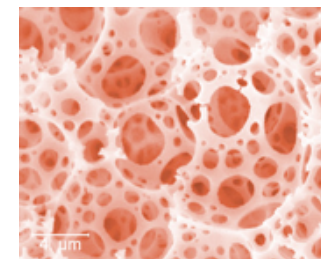
- Empirical



- Geometrical



- Semi phenomenological



Idea: Fit model with experiments

- Experiments by Delany and Bazley (1970)
- Extensive investigation on fibrous absorbents (wide range of commercial)
- Porosity close to 1
- Measured values expressed in terms of power series

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$$A, B, C, D = \alpha X^\beta \quad X?$$

Which parameter can catch most of the physics ?

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$$\rho_0 \frac{\partial v}{\partial t} + \sigma v = - \nabla p$$

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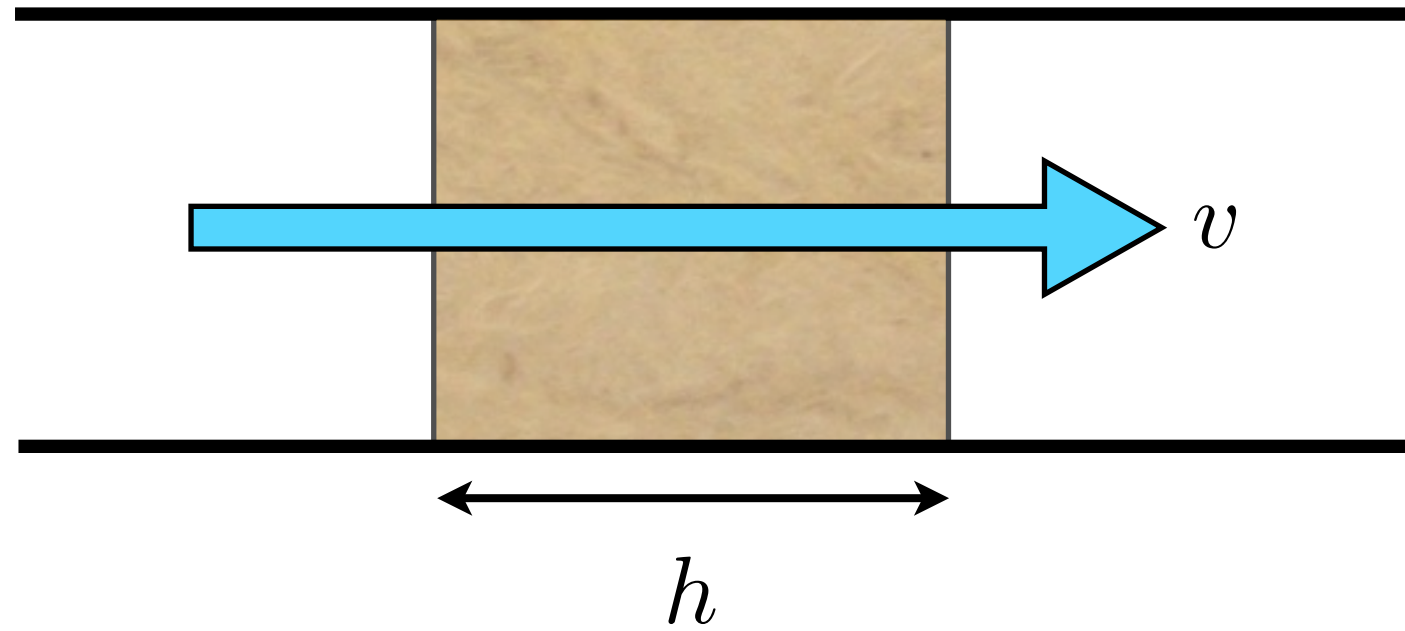
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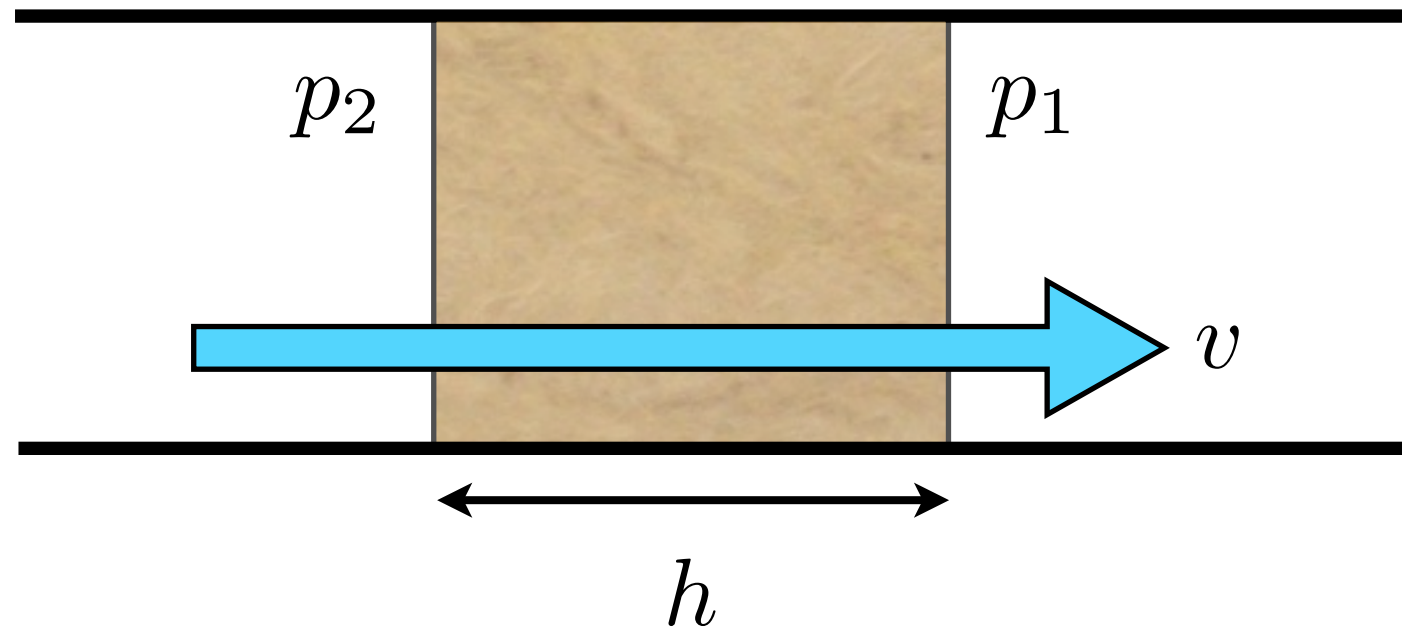
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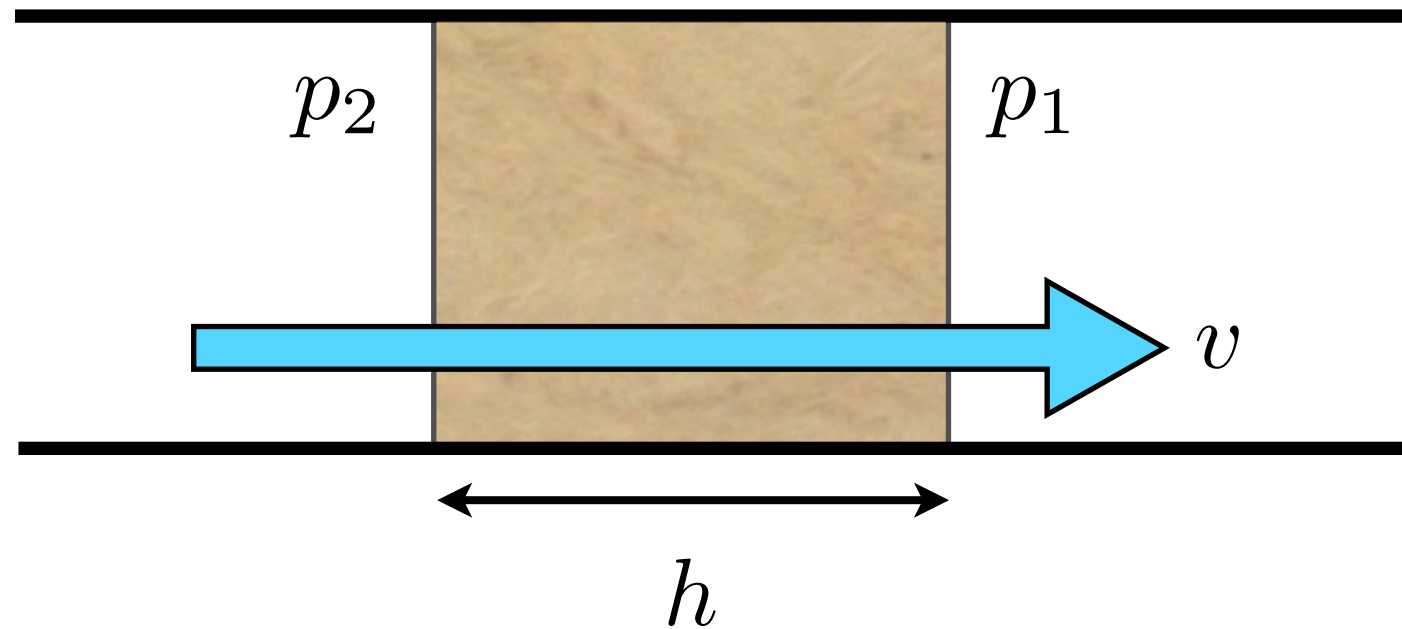
$$(j \rho_0 2\pi f + \sigma) v = - \nabla p \quad X = \frac{f}{\sigma}$$

Flow resistivity



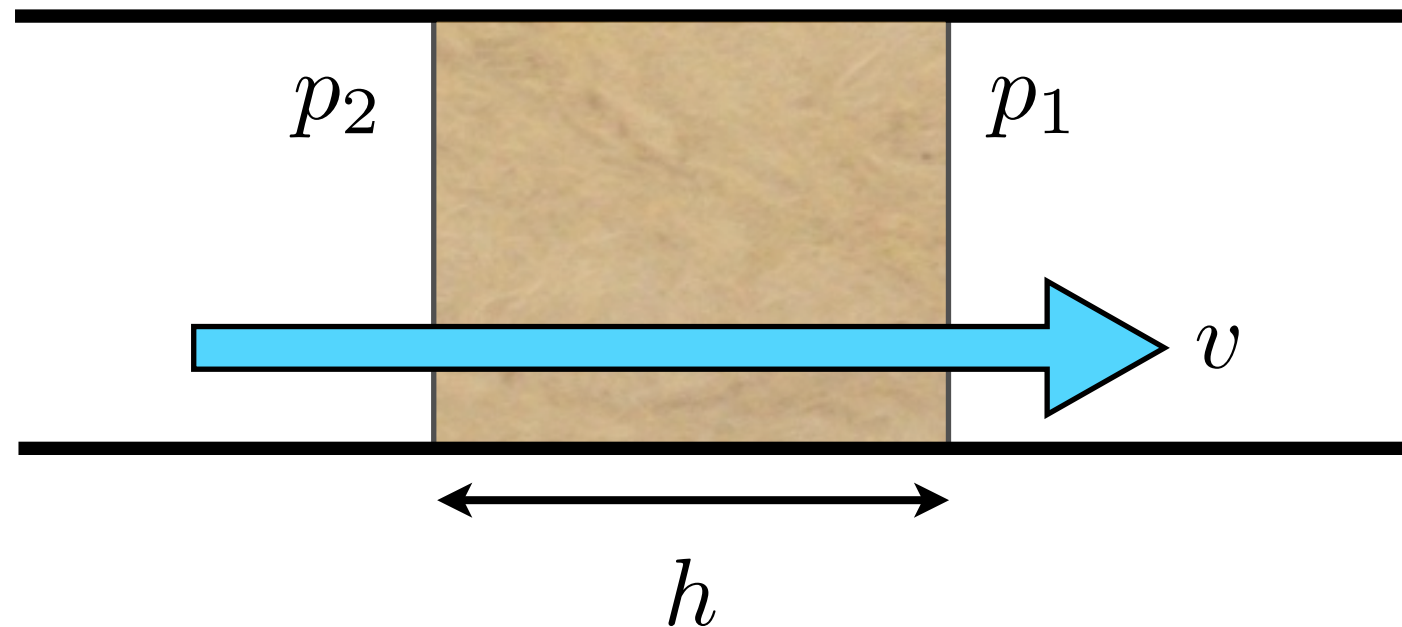
Flow resistivity





$$\sigma = \frac{p_2 - p_1}{v \times h}$$

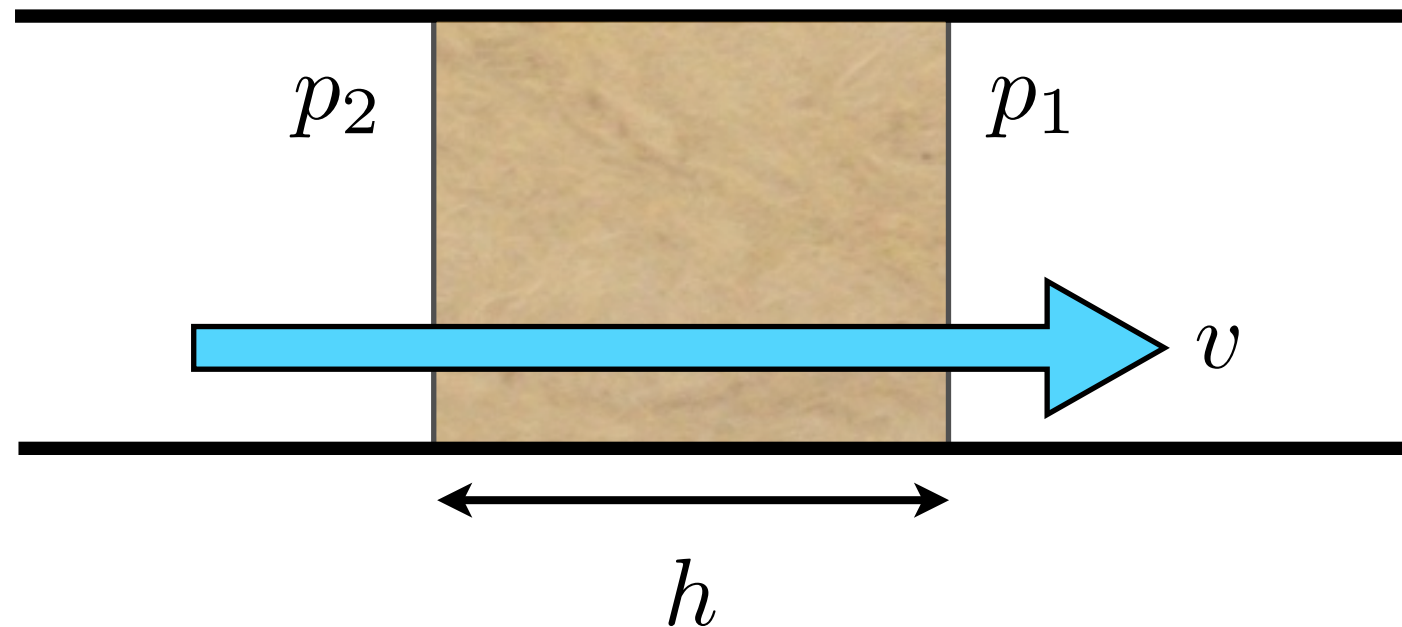
Unit: $N m^{-4} s$



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- ✓ Values between 1 000 and 100 000 $N m^{-4} s$
- ✓ Standard: ASTM C-522, ISO 9053
- ✓ Sensitive to anisotropy (Tensor)



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Darcy law

$$\mathbf{v}_{eq} = -\frac{k_0}{\eta} \nabla p$$
$$k_0 = \frac{\eta}{\sigma}$$

Viscous permeability

$$Z_{eq} = \rho_0 c_0 (1 + A + j B), \quad \delta_{eq} = \frac{\omega}{c_0} (1 + C + j D)$$

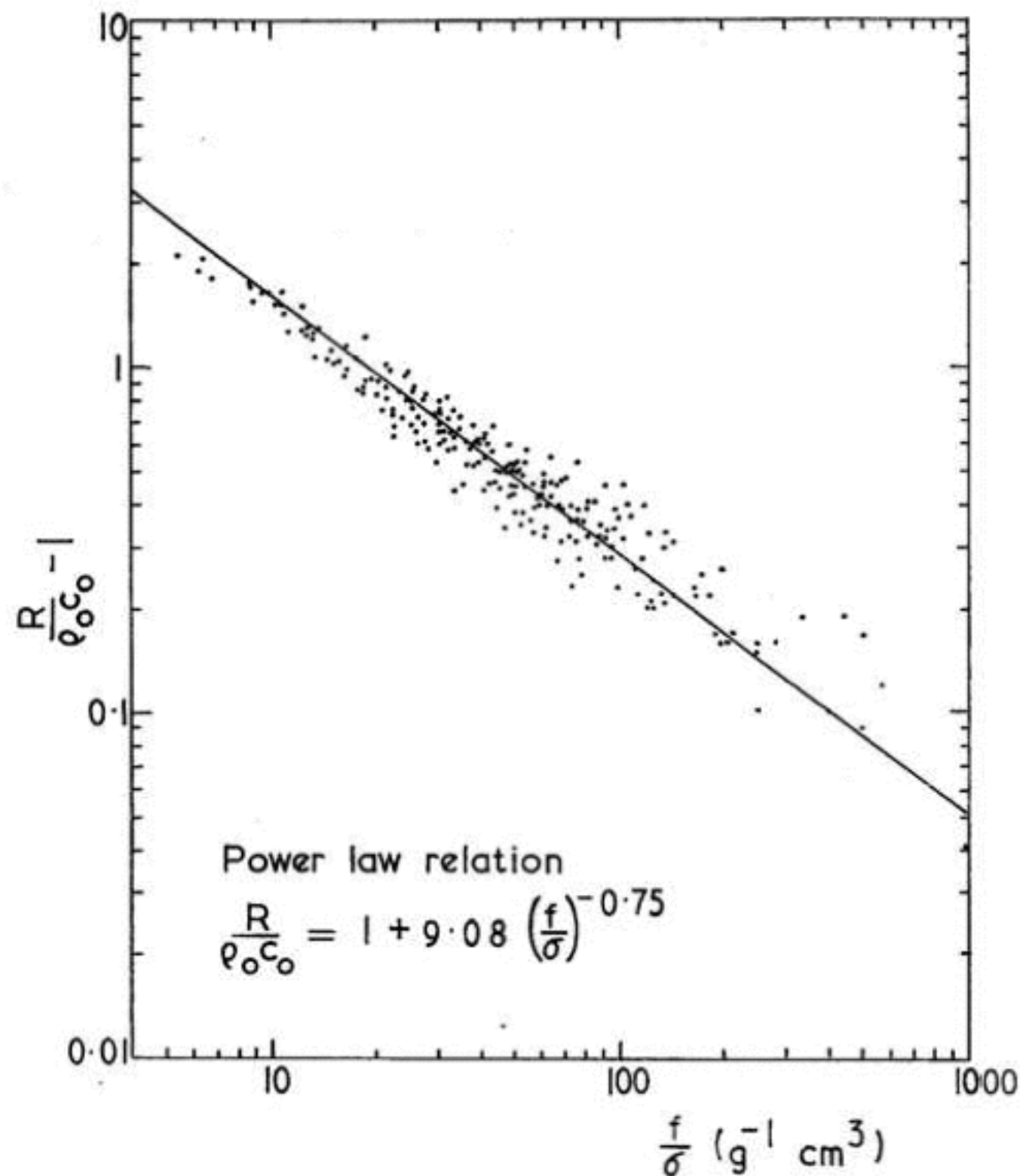
$$A = a \left(\frac{f}{\sigma} \right)^b, \quad B = c \left(\frac{f}{\sigma} \right)^d, \quad C = q \left(\frac{f}{\sigma} \right)^r, \quad D = p \left(\frac{f}{\sigma} \right)^s$$

$$Z_{eq} = \rho$$

$$A = a \left(\frac{f}{\sigma} \right)^t$$

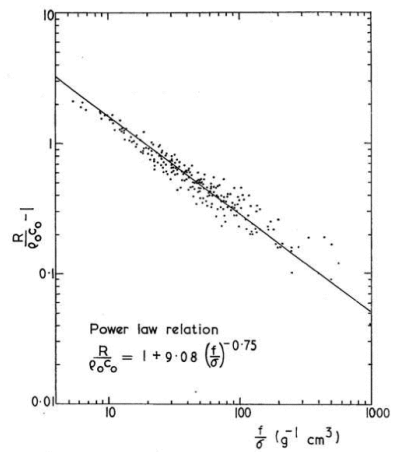
$$B = b \left(\frac{f}{\sigma} \right)^u + j D)$$

$$C = p \left(\frac{f}{\sigma} \right)^s$$



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$$Z_{eq}(\omega) = \rho_0 c_0 \left[1 + 9.08 \left(\frac{f}{\sigma} \right)^{-0.75} - j 11.9 \left(\frac{f}{\sigma} \right)^{-0.73} \right]$$

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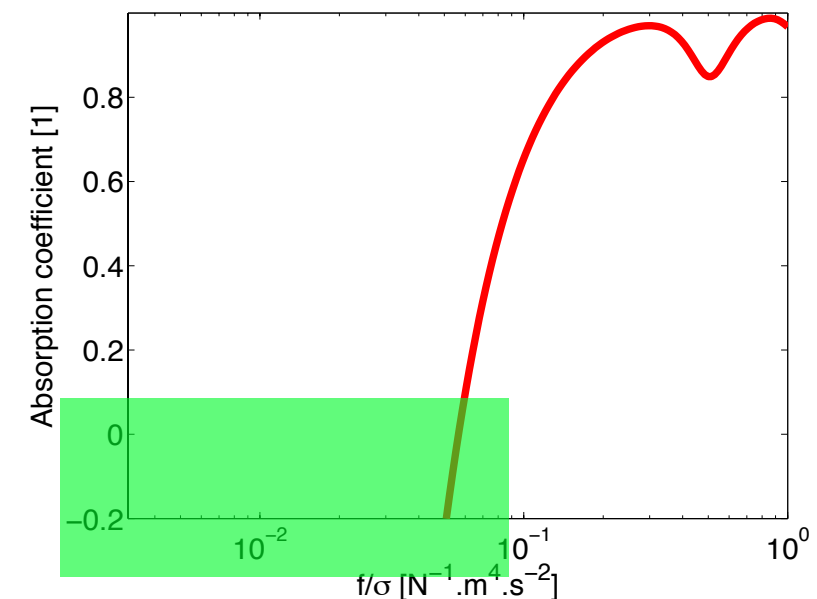
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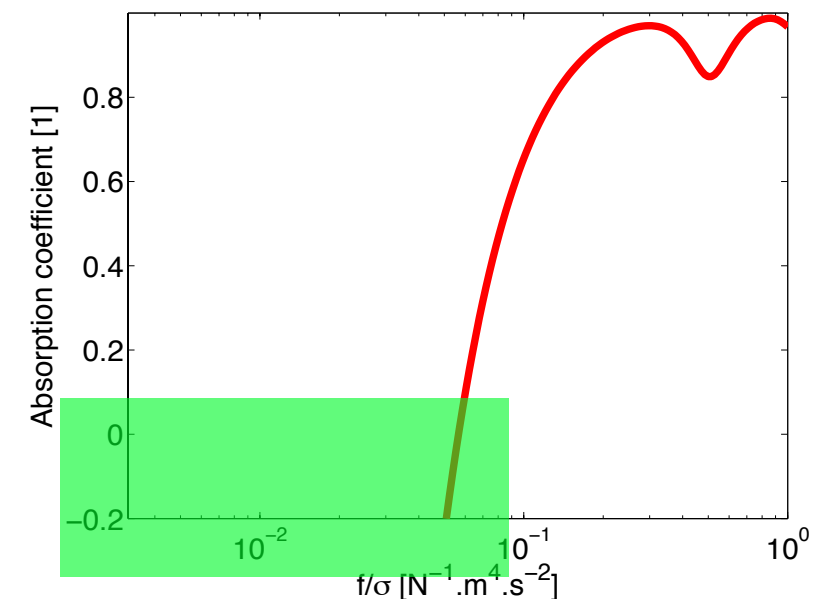


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Adimensional parameter $X = \frac{\rho_0 f}{\sigma}$



$$Z_{eq} = \rho_0 c_0 \left[1 + 0.0571 X^{-0.754} - j0.087 X^{-0.732} \right]$$

$$\delta_{eq} = \frac{\omega}{c_0} \left[1 + 0.0978 X^{-0.700} - j0.189 X^{-0.595} \right]$$

Refinement of DB model including causality in the fitting process (constraint optimisation)

$$Z_{eq} = \rho_0 c_0 (1 + A + j B), \quad \delta_{eq} = \frac{\omega}{c_0} (1 + C + j D)$$
$$A(f) = a f^b, \quad B(f) = c f^d, \quad C(f) = q f^r, \quad D(f) = p f^s$$

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Positive real part of the impedance function

$$d = b, \quad c = -a \cot \left(\frac{(b+1)\pi}{2} \right), \quad -1 < b < 0, \quad r = s, \quad p = -q \tan \left(\frac{r\pi}{2} \right)$$

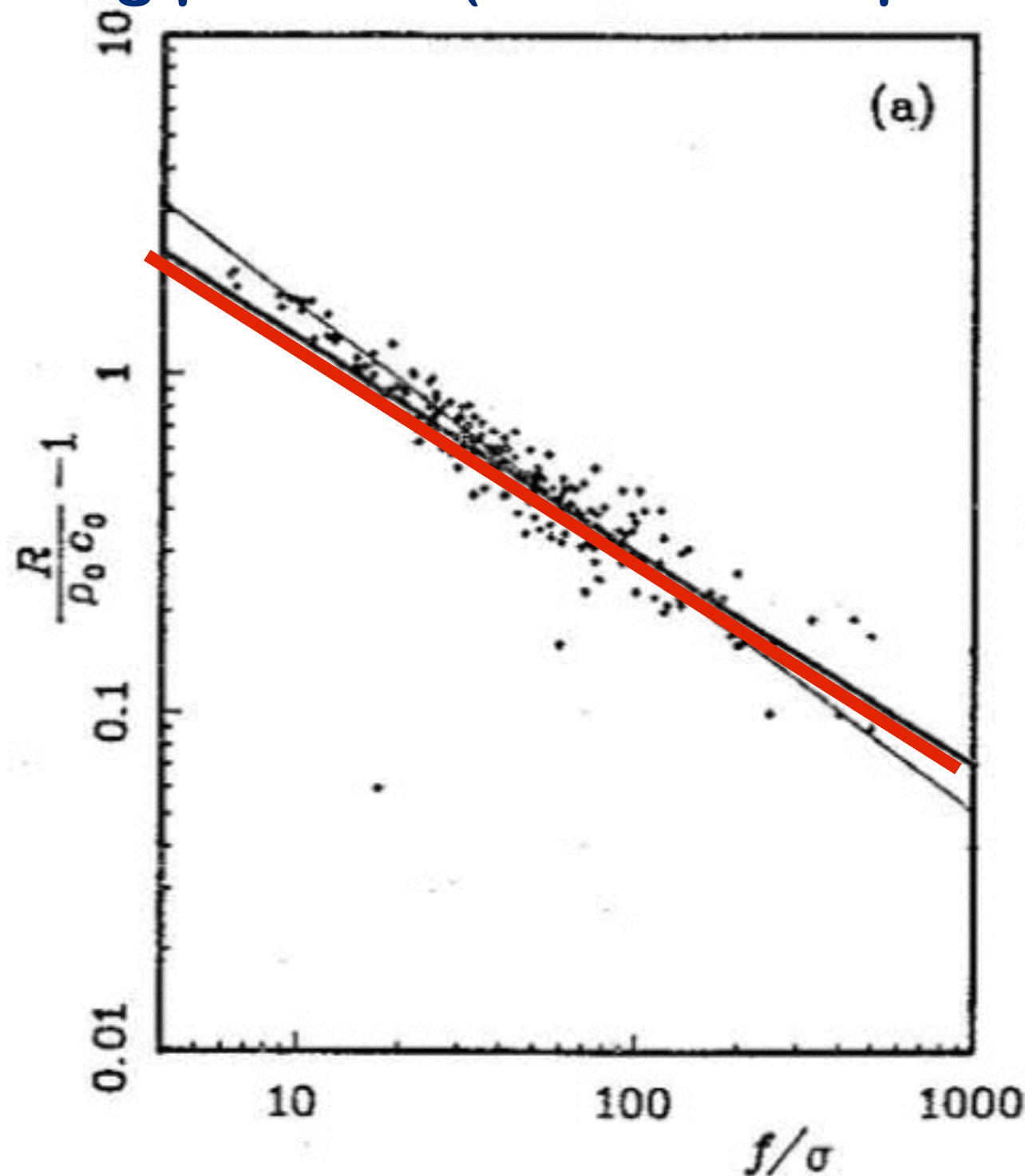
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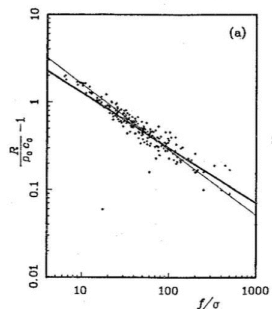
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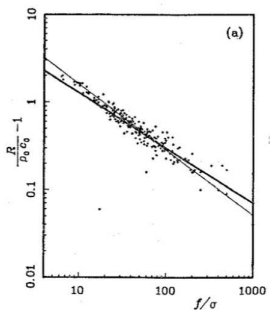
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$$Z_{eq} = \rho_0 c_0 \left[1 + 0.0785 X^{-0.632} - j 0.120 X^{-0.632} \right],$$

$$\delta_{eq} = \frac{\omega}{c_0} \left[1 + 0.122 X^{-0.618} - j 0.180 X^{-0.618} \right].$$



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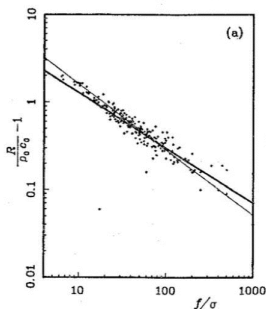
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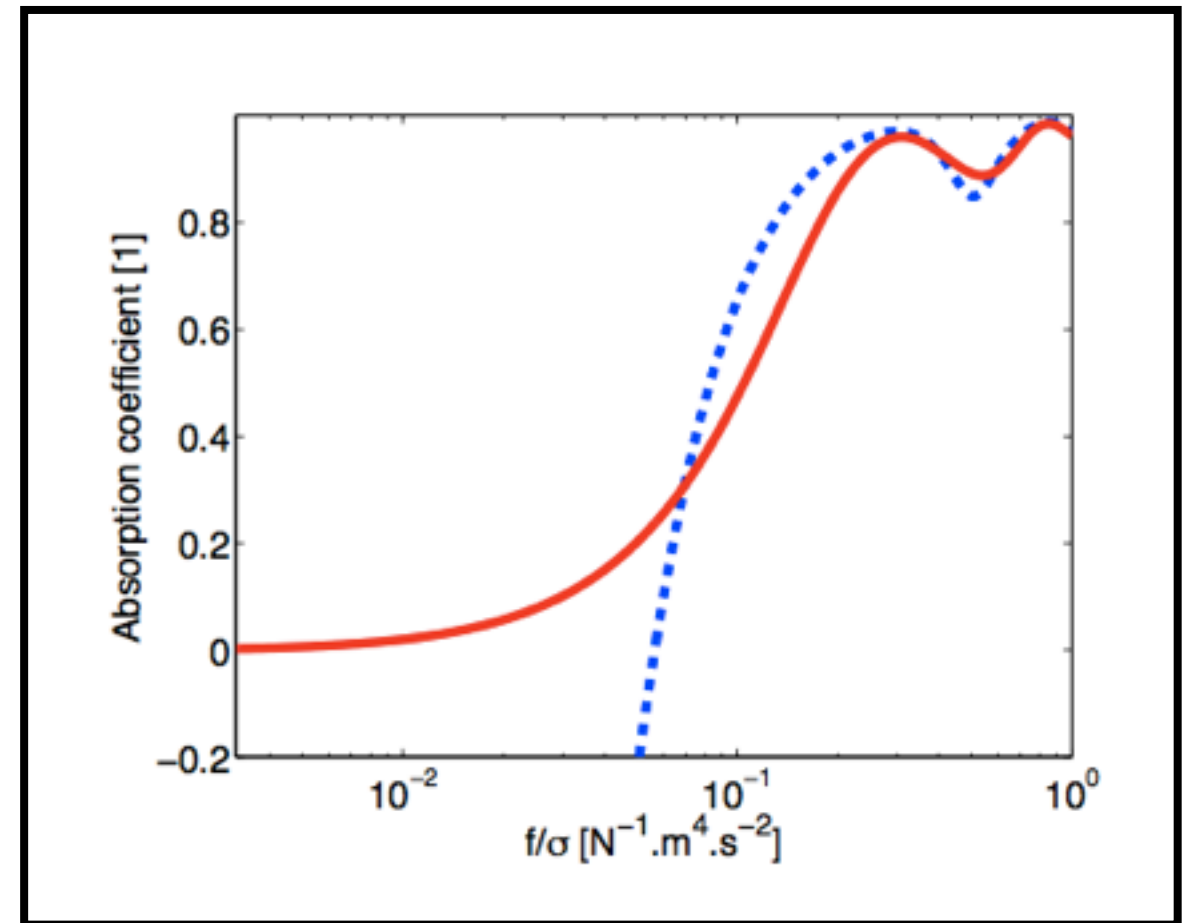
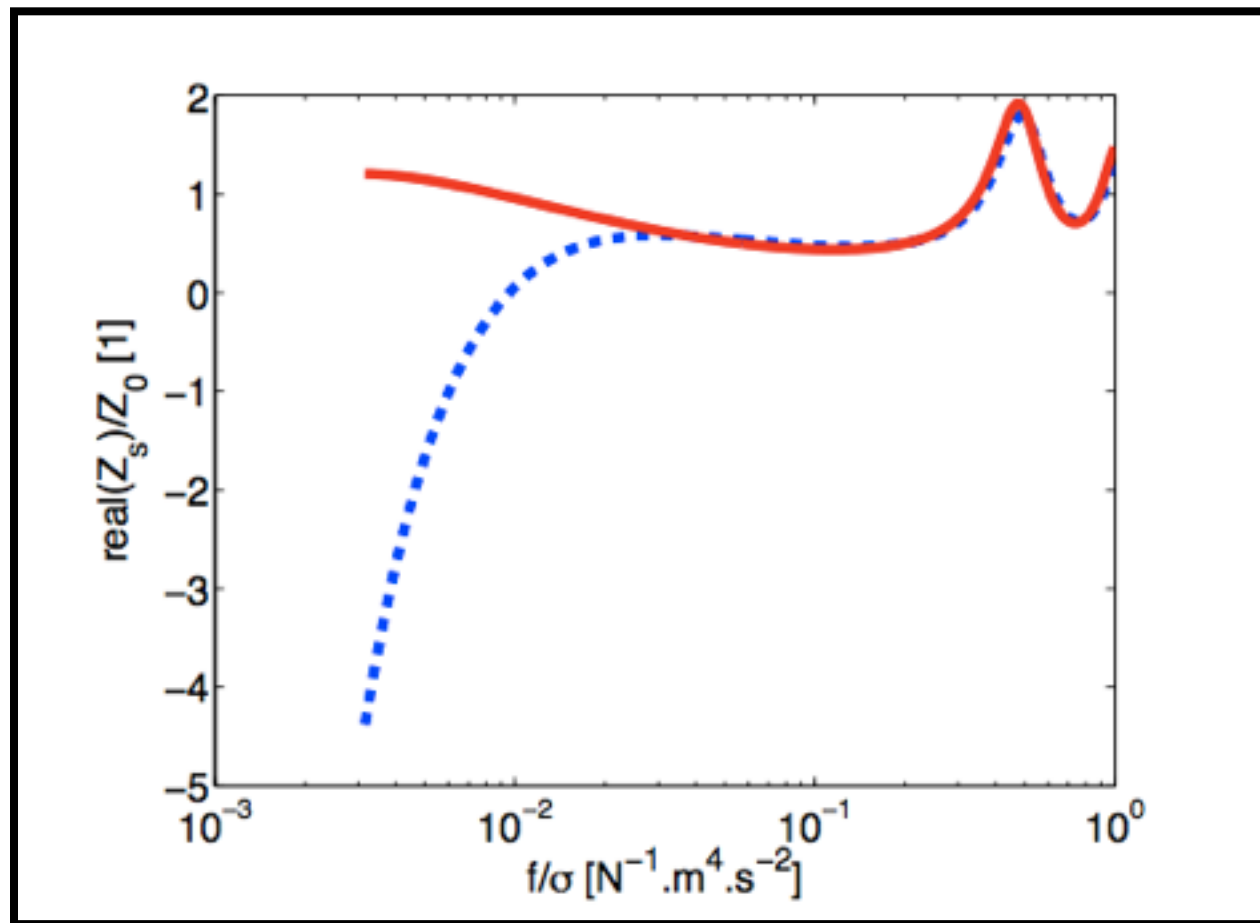
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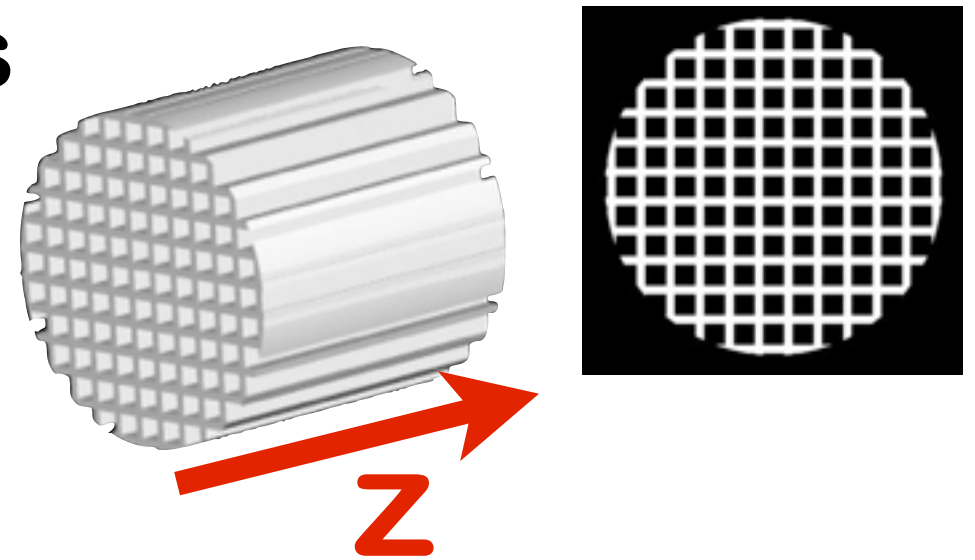


To take porosity into account: $X \longrightarrow X' = \frac{\rho_0 f}{\phi \sigma}$

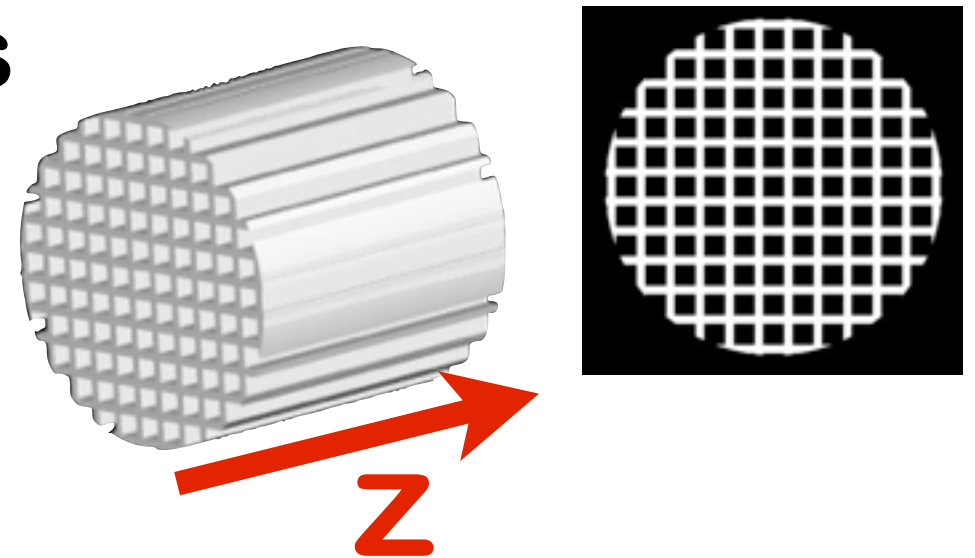
$$\sigma = 10\,000 N.s.m^{-4}$$



- Perfect microscopical geometry
- Analytical expressions
- Propagation along the tube direction (z)
- Cylinders of different shapes
 - Circular
 - Rectangles
 - Slits

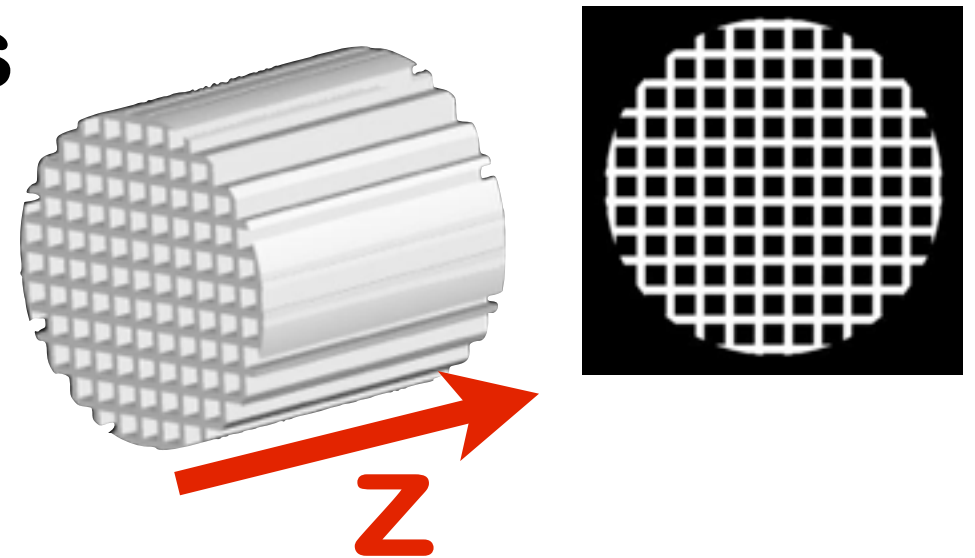


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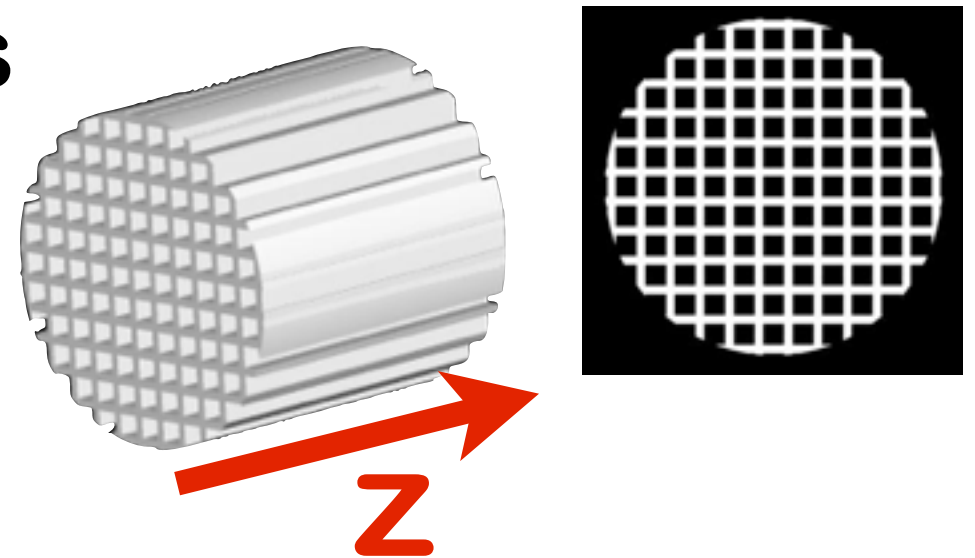
$$\underbrace{\vec{v}(x, y, z)}_{\mathbf{x}_\perp} = [\vec{v}_\perp(\mathbf{x}_\perp, \omega) + v_z(\mathbf{x}_\perp, \omega)\vec{e}_z] e^{i\delta z}$$

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Differential operator $\nabla^2 = \nabla_\perp^2 + \frac{\partial^2}{\partial z^2}$

$$\nabla_{\perp}^2 \vec{v}_z - \frac{j\omega}{\nu} \vec{v}_z = \frac{1}{\eta} \frac{\partial p}{\partial z}$$

$$v_z|_{\Gamma} = \mathbf{0}$$

Navier-Stokes equation

$$\nabla_{\perp}^2 \tau - \frac{j\omega\gamma}{\nu'} \tau = -\frac{j\omega}{\kappa} p$$

$$\tau|_{\Gamma} = \mathbf{0}$$

Thermal equation

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Thermal equation

General form of both problems

$$\nabla_{\perp}^2 \psi - \frac{j\omega}{\nu} \psi = -\frac{j\omega}{\nu}, \quad \psi = 0 \text{ on the boundary}$$

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Resolution of this problem for each geometry $\longrightarrow \psi(\omega)$

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$$v_z(\omega) = -\frac{1}{j\omega\rho_0} \frac{\partial p}{\partial z} \psi(\omega)$$

$$\tau(\omega) = -\frac{\nu'}{\gamma\kappa} \psi(Pr\omega) p$$

$$\gamma = \frac{C_P}{C_v}$$

$$Pr = \frac{\eta C_P}{\kappa} \text{ **Prandtl number**}$$

$$v_z(\mathbf{x}_\perp, \omega) = -\frac{1}{j\omega\rho_0} \psi(\mathbf{x}_\perp, \omega) \frac{\partial p}{\partial z}$$

$$v_z(\underline{\mathbf{x}}_{\perp}, \omega) = -\frac{1}{j\omega\rho_0} \psi(\underline{\mathbf{x}}_{\perp}, \omega) \frac{\partial p}{\partial z}$$

Averaging on the tube section

Constant on the section

$$j\omega\rho_0 \langle v_z(\mathbf{x}_{\perp}, \omega) \rangle_S = - \langle \psi(\mathbf{x}_{\perp}, \omega) \rangle_S \frac{\partial p}{\partial z}$$

$$v_z(\underline{\mathbf{x}}_{\perp}, \omega) = -\frac{1}{j\omega\rho_0} \psi(\underline{\mathbf{x}}_{\perp}, \omega) \frac{\partial p}{\partial z}$$

Averaging on the tube section

Constant on the section

$$j\omega\rho_0 \langle v_z(\underline{\mathbf{x}}_{\perp}, \omega) \rangle_S = - \langle \psi(\underline{\mathbf{x}}_{\perp}, \omega) \rangle_S \frac{\partial p}{\partial z}$$

$$\tilde{\rho}_{eq}(\omega) = \frac{\rho_0}{\langle \psi(\underline{\mathbf{x}}_{\perp}, \omega) \rangle_S}$$

Apparent effective density

Motion equation

$$j\omega\tilde{\rho}_{eq}\mathbf{v}^{eq} = -\nabla p$$

$$\tau(\mathbf{x}_{\perp}, \omega) = -\frac{\nu'}{\gamma\kappa} \psi(\mathbf{x}_{\perp}, Pr \omega) p$$

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State equation averaged

$$\langle \rho \rangle_S = \frac{\rho_0}{P_0} p - \frac{\rho_0}{T_0} \langle \tau \rangle_S$$

Mass conservation averaged

$$\langle \rho \rangle_S = -\rho_0 \langle \nabla \cdot \vec{u} \rangle_S$$

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$$-\langle \nabla \cdot \vec{u} \rangle = \left(\frac{1}{P_0} - \frac{1}{T_0} \frac{\nu'_0}{\kappa \gamma} \langle \psi(Pr \omega) \rangle \right) p$$

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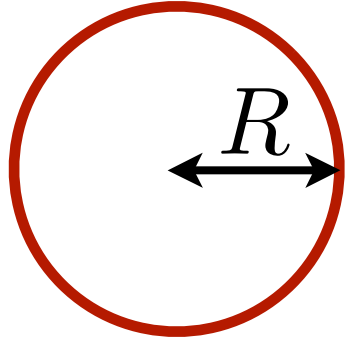
$$\langle \tau \rangle_S = -\frac{\nu'}{\gamma \kappa} \langle \psi(Pr \omega) \rangle_S p$$

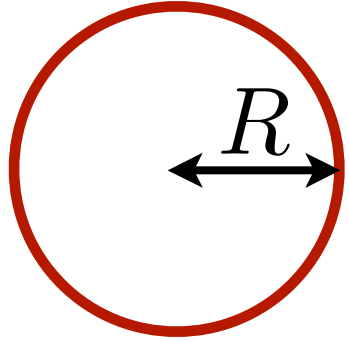
$$p = -\tilde{K}_{eq} \nabla \cdot \mathbf{u}^{eq}$$

$$\tilde{K}_{eq}(\omega) = \frac{\gamma P_0}{\gamma - (\gamma - 1) \langle \psi(\mathbf{x}_{\perp}, Pr \omega) \rangle_S}$$

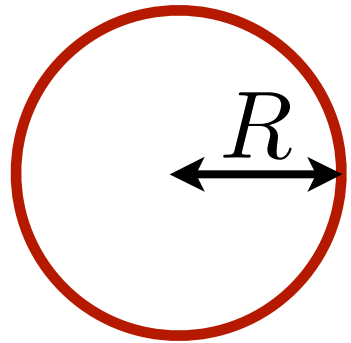
Apparent effective compressibility

Case of circular tubes





$$\nabla_{\perp}^2 \psi - \frac{j\omega}{\nu} \psi = -\frac{j\omega}{\nu}$$

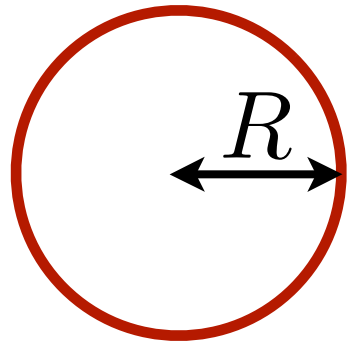


Bessel equation
$$\frac{d^2\psi(r)}{dr^2} + \frac{1}{r} \frac{d\psi(r)}{dr} - \frac{j}{\delta_v^2} \psi(r) = -\frac{j}{\delta_v^2}$$

Boundary condition $\psi(R) = 0$

$$\delta_v = \sqrt{\frac{2\nu}{\omega}}$$

Viscous length



Bessel equation
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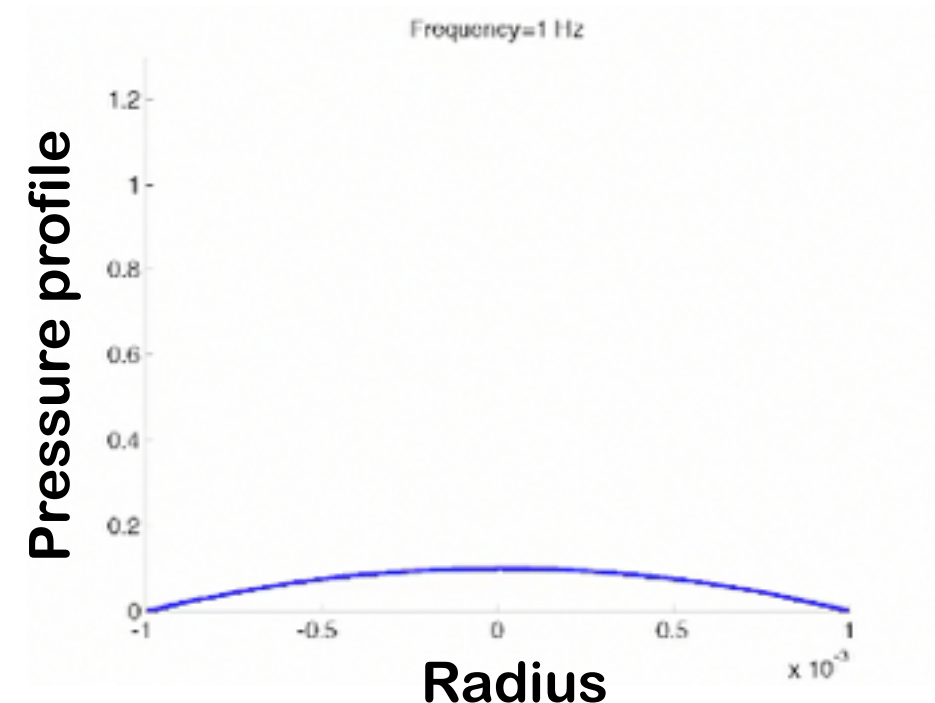
Viscous length

Useful result

Bessel function $J_0(x)$, solution of
$$\frac{d^2y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + y = 0$$

Solution of the generic problem

$$\psi(r, \omega) = 1 - \frac{J_0\left(\sqrt{-2j} \frac{r}{\delta_v}\right)}{J_0\left(\sqrt{-2j} \frac{R}{\delta_v}\right)}$$



$$\langle \psi(\omega) \rangle_s = 1 - \frac{2}{s\sqrt{-j}} \frac{J_1(s\sqrt{-j})}{J_0(s\sqrt{-j})}$$

Relevant parameter

$$s = R\sqrt{\frac{\omega}{\nu}}$$

$$\langle \psi(\omega) \rangle_s = 1 - \frac{2}{s\sqrt{-j}} \frac{J_1(s\sqrt{-j})}{J_0(s\sqrt{-j})}$$

Relevant parameter

$$s = R\sqrt{\frac{\omega}{\nu}}$$

$$\tilde{\rho}_{eq}(\omega) = \frac{\rho_0}{\langle \psi(\omega) \rangle_s} = \rho_0 \tilde{\alpha}(\omega)$$

Dynamic tortuosity $\tilde{\alpha}(\omega) = \frac{1}{\langle \psi(\omega) \rangle_s}$

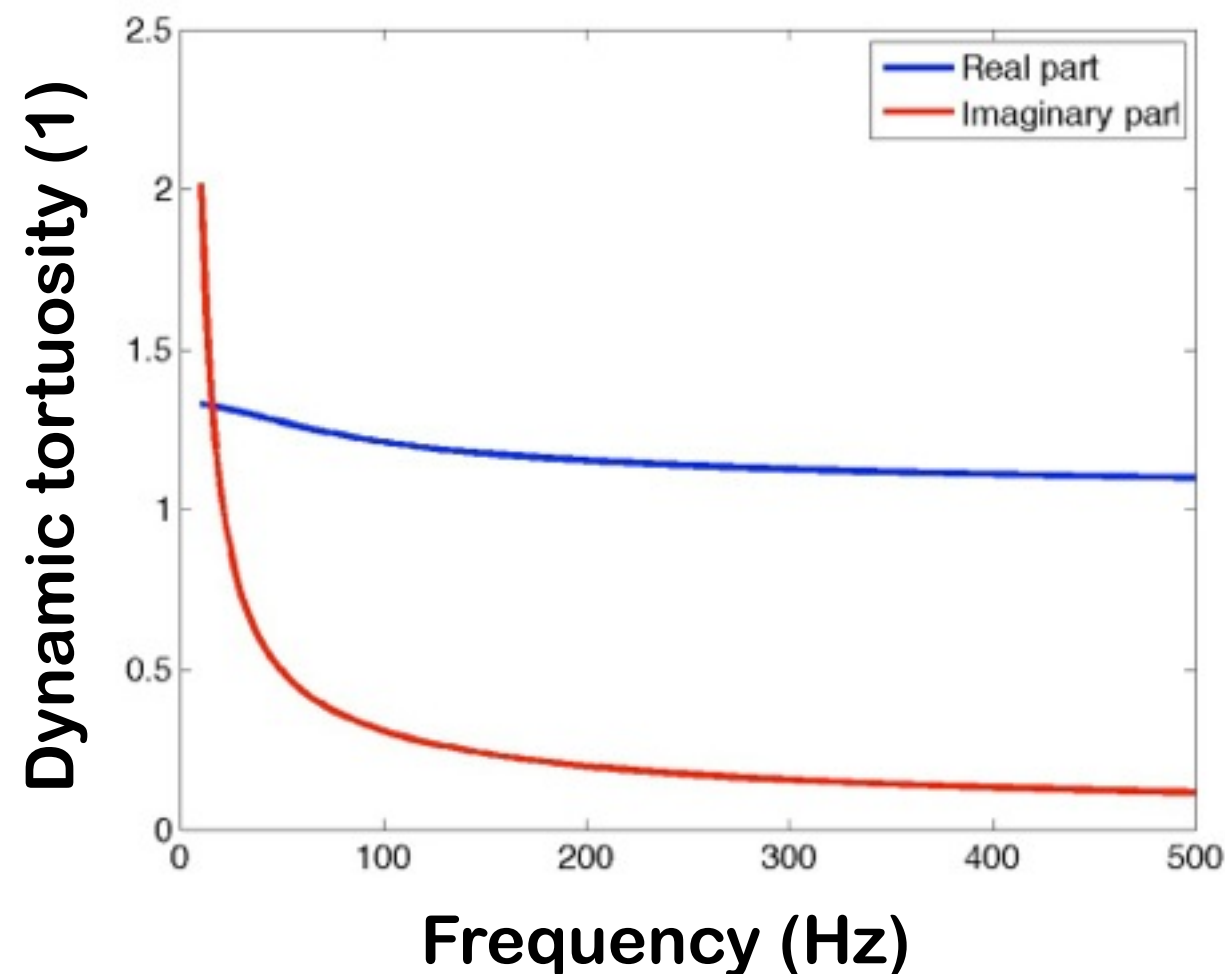
$$\langle \psi(\omega) \rangle_S = 1 - \frac{2}{s\sqrt{-j}} \frac{J_1(s\sqrt{-j})}{J_0(s\sqrt{-j})}$$

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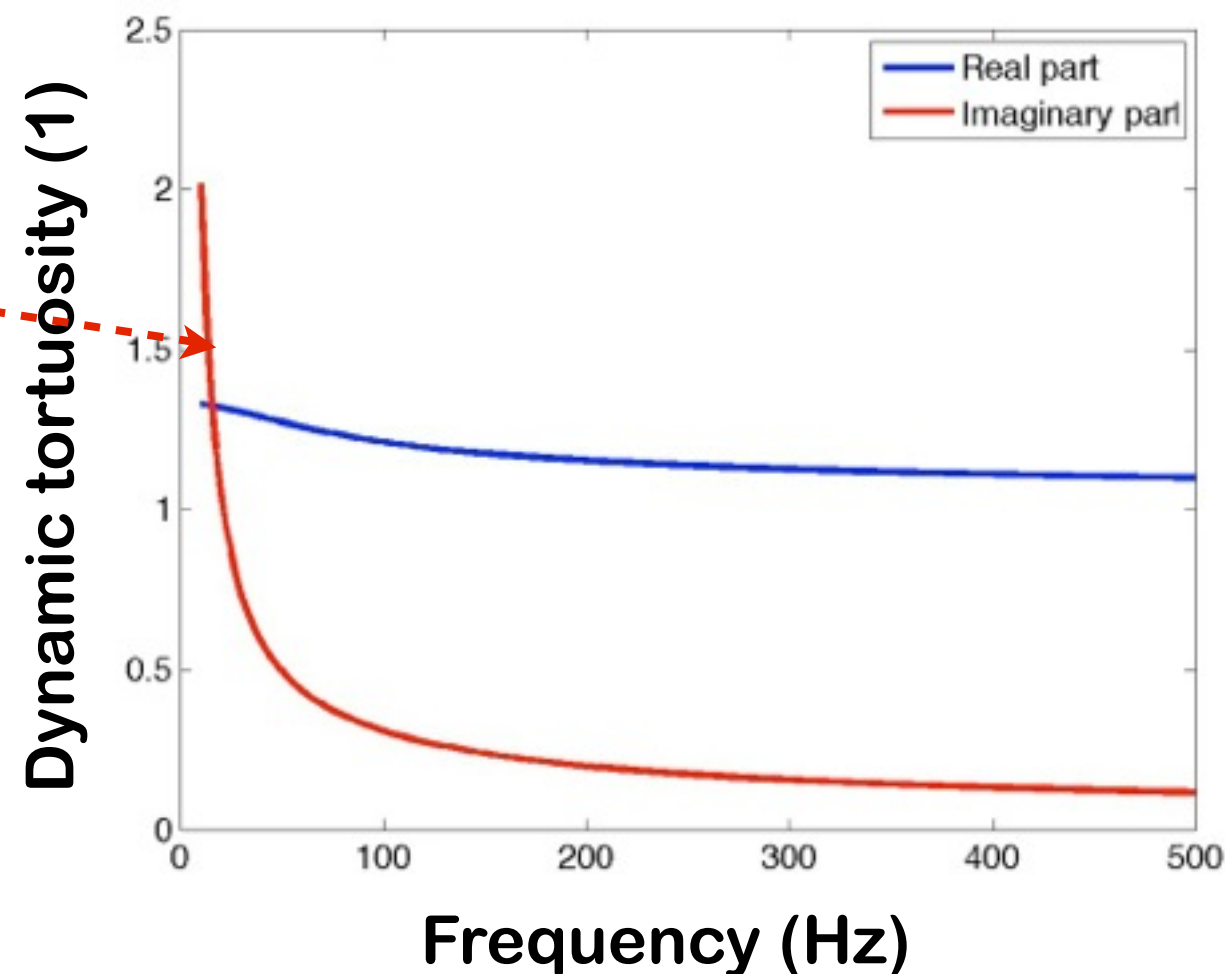
$$\tilde{\rho}_{eq}(\omega) = \frac{\rho_0}{\langle \psi(\omega) \rangle_S} = \rho_0 \tilde{\alpha}(\omega)$$

Dynamic tortuosity $\tilde{\alpha}(\omega) = \frac{1}{\langle \psi(\omega) \rangle_S}$

$$\omega \rightarrow 0$$

$$\Re(\tilde{\alpha}) \ll \Im(\tilde{\alpha})$$

$$\Im(\tilde{\alpha}) \rightarrow \infty$$



$$\langle \psi(\omega) \rangle_S = 1 - \frac{2}{s\sqrt{-j}} \frac{J_1(s\sqrt{-j})}{J_0(s\sqrt{-j})}$$

Relevant parameter

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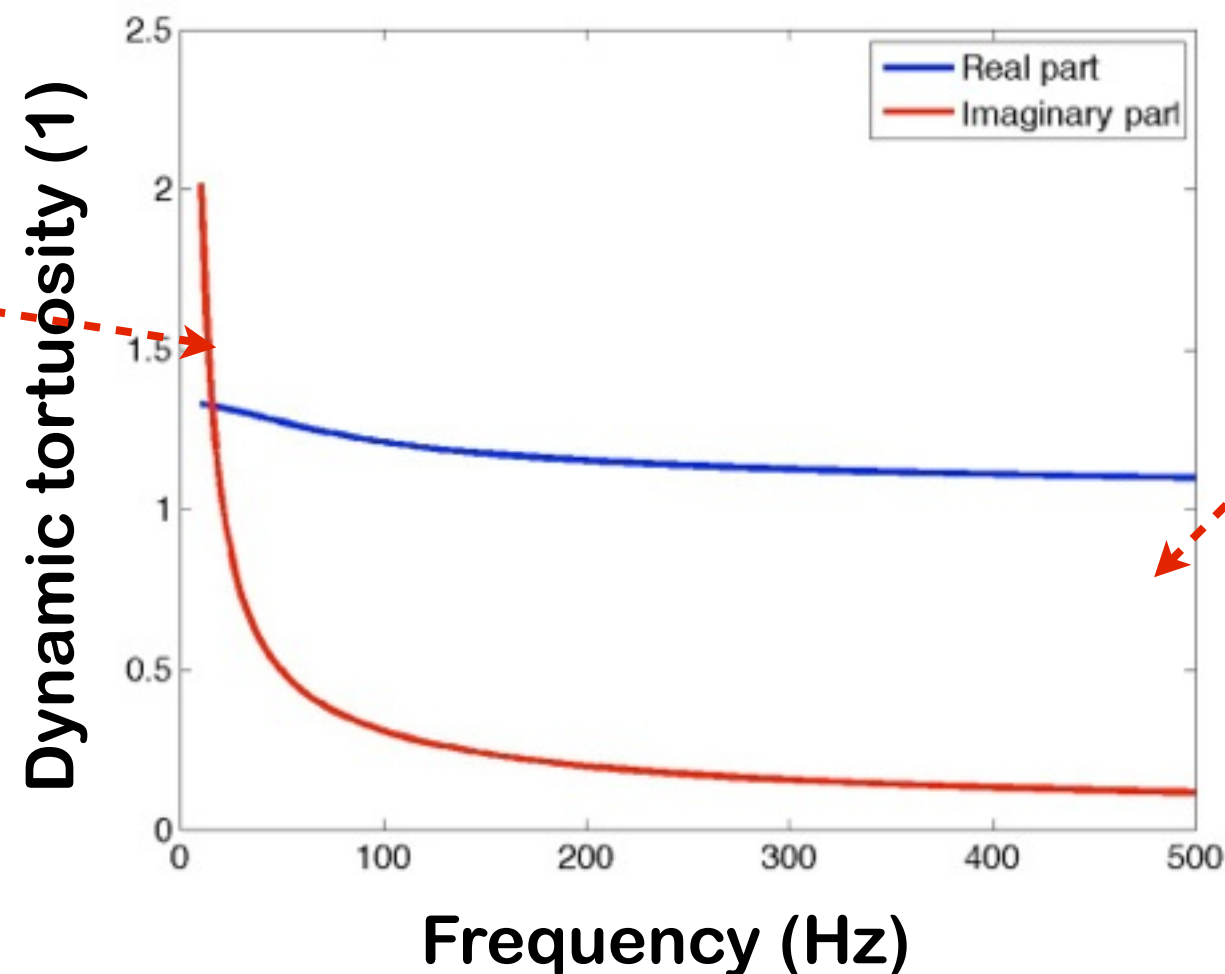
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Dynamic tortuosity $\tilde{\alpha}(\omega) = \frac{1}{\langle \psi(\omega) \rangle_S}$

$$\omega \rightarrow 0$$

$$\Re(\tilde{\alpha}) \ll \Im(\tilde{\alpha})$$

$$\Im(\tilde{\alpha}) \rightarrow \infty$$



$$\omega \rightarrow \infty$$

$$\Re(\tilde{\alpha}) \rightarrow 1$$

$$\Im(\tilde{\alpha}) \rightarrow 0$$

$$\lim_{\omega \rightarrow 0} \tilde{\alpha}(\omega) = \frac{8\nu}{j\omega R^2} + \underbrace{\frac{4}{3}}_{\alpha_0} + O(j\omega)$$

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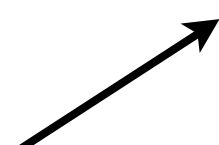
Imaginary part: dominant term

$$\lim_{\omega \rightarrow 0} \tilde{\alpha}(\omega) = \frac{8\nu}{j\omega R^2} + \underbrace{\frac{4}{3}}_{\alpha_0} + O(j\omega)$$

Imaginary part: dominant term

Motion equation

$$\rho_0 \left(\frac{8\nu}{R^2} + \alpha_0 j\omega \right) \mathbf{v}^{eq} = -\nabla p$$

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Imaginary part: dominant term

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$$\rho_0 \left(\frac{8\nu}{R^2} + \cancel{\alpha_0 j\omega} \right) \mathbf{v}^{eq} = -\nabla p$$

Diffusion (Darcy) law

$$\mathbf{v}_{eq} = -\frac{k_0}{\eta} \nabla p$$

$$\lim_{\omega \rightarrow 0} \tilde{\alpha}(\omega) = \frac{8\nu}{j\omega R^2} + \underbrace{\frac{4}{3}}_{\alpha_0} + O(j\omega)$$

Imaginary part: dominant term

Motion equation

$$\rho_0 \left(\frac{8\nu}{R^2} + \cancel{\alpha_0 j\omega} \right) \mathbf{v}^{eq} = -\nabla p$$

$$k_0 = \frac{R^2}{8} \quad \text{Diffusion (Darcy) law} \quad \mathbf{v}_{eq} = -\frac{k_0}{\eta} \nabla p$$

Viscous permeability

$$\lim_{\omega \rightarrow 0} \tilde{\alpha}(\omega) = \frac{8\nu}{j\omega R^2} + \underbrace{\frac{4}{3}}_{\alpha_0} + O(j\omega)$$

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Viscous permeability

Value of flow resistivity

$$\sigma = \frac{8\eta}{R^2}$$

Asymptotic development of Bessel functions

$$\lim_{x \rightarrow \infty} \frac{J_1(x)}{J_0(x)} = j$$

$$\lim_{\omega \rightarrow \infty} \tilde{\alpha}(\omega) = \underbrace{1}_{\alpha_\infty} \underbrace{-\frac{2}{R} \sqrt{\frac{\nu}{j\omega}}}_{(1-j) \frac{\delta_v}{R}}$$

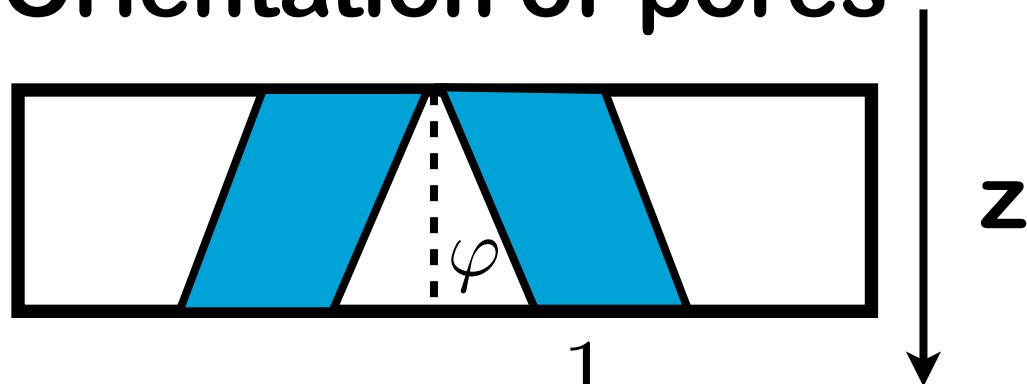
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Geometric tortuosity

Orientation of pores



$$\alpha_\infty = \frac{1}{\cos^2(\varphi)}$$

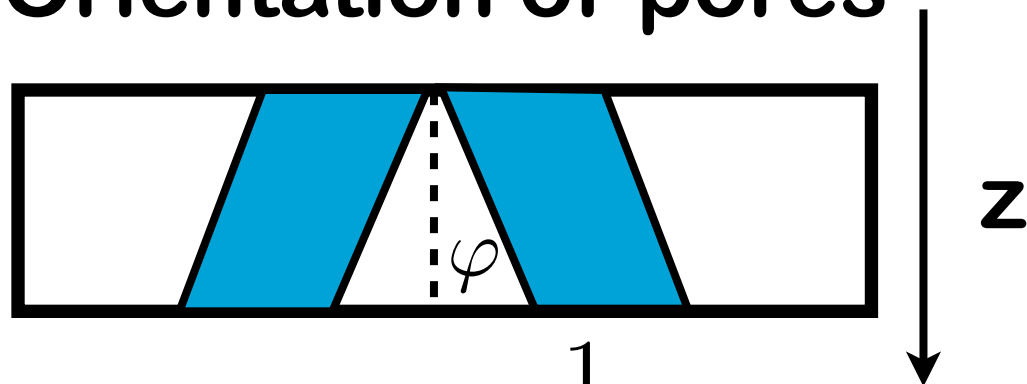
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Orientation of pores



$$\alpha_\infty = \frac{1}{\cos^2(\varphi)}$$

$$(j\omega)^{-1/2}$$

Low decrease (HF)

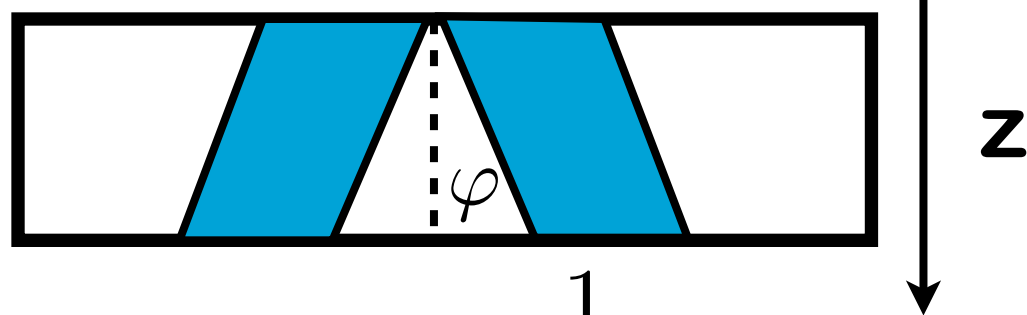
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Geometric tortuosity

Orientation of pores



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$(j\omega)^{-1/2}$

Low decrease (HF)

Concept of Viscous Characteristic length

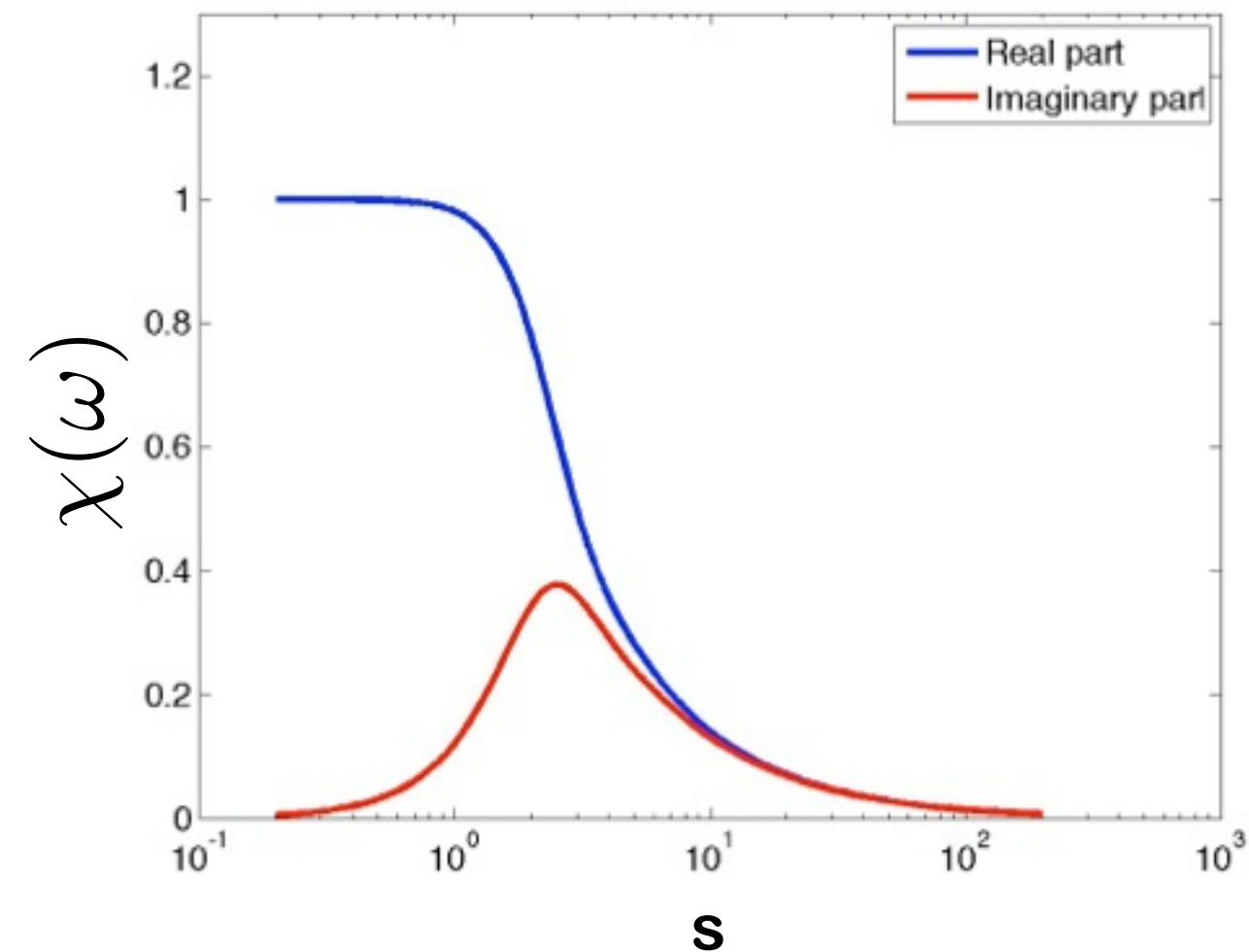
$$\Lambda = R$$

Relaxation function
(Analogy with dielectric
relaxation)

$$\chi(\omega) = 1 - \frac{\alpha_{\infty}}{\tilde{\alpha}(\omega)}$$

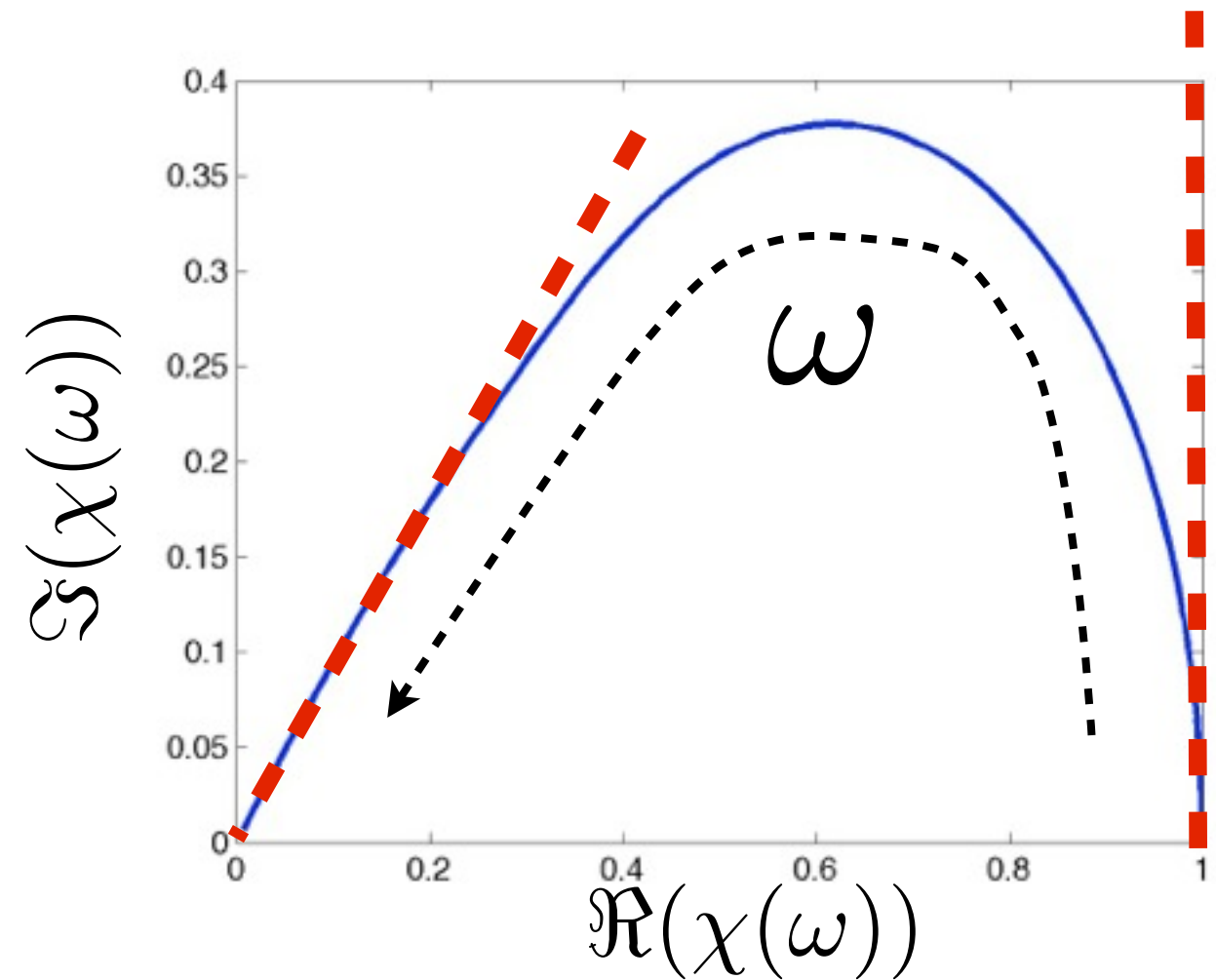
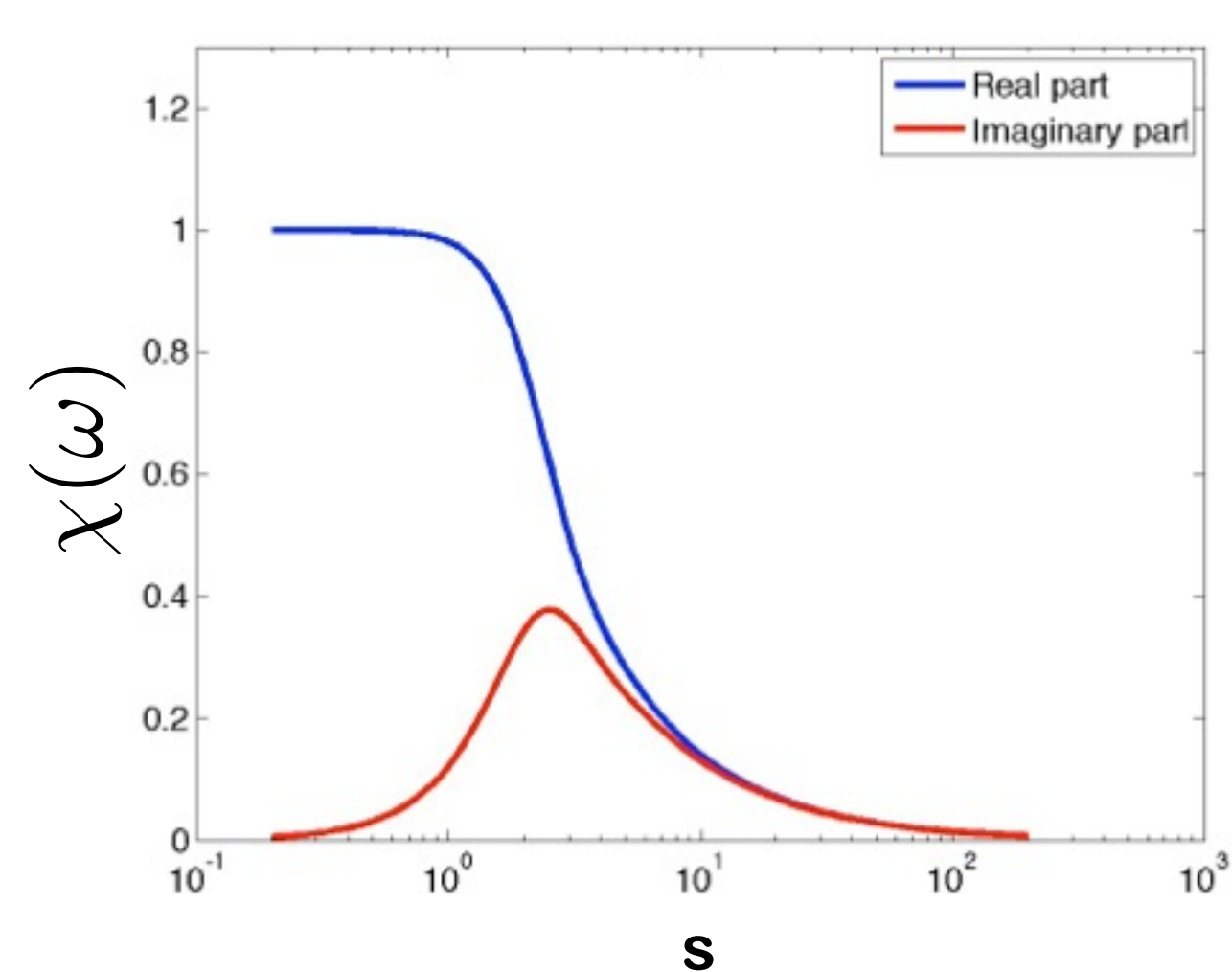
Relaxation function (Analogy with dielectric relaxation)

$$\chi(\omega) = 1 - \frac{\alpha_{\infty}}{\tilde{\alpha}(\omega)}$$



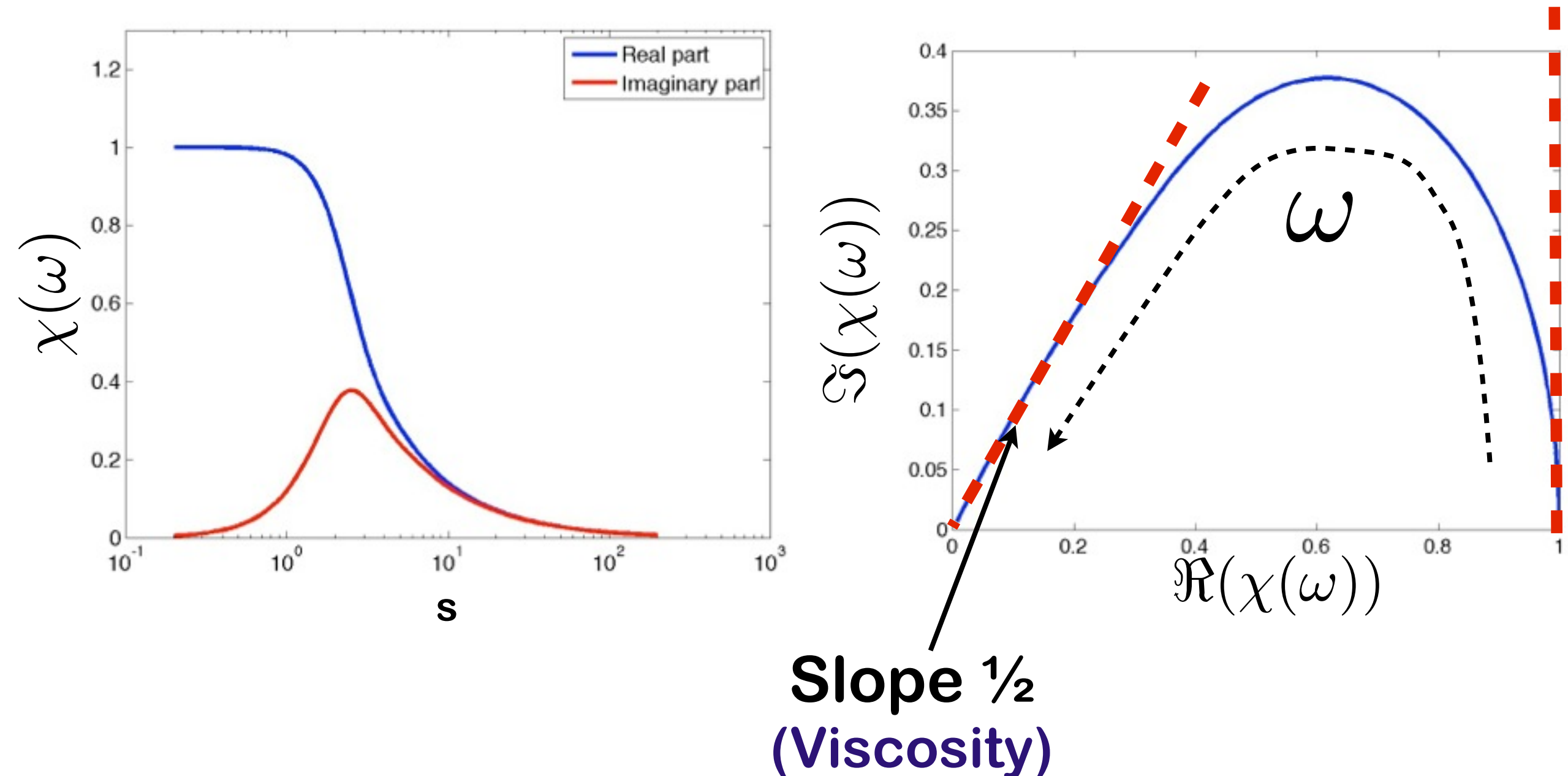
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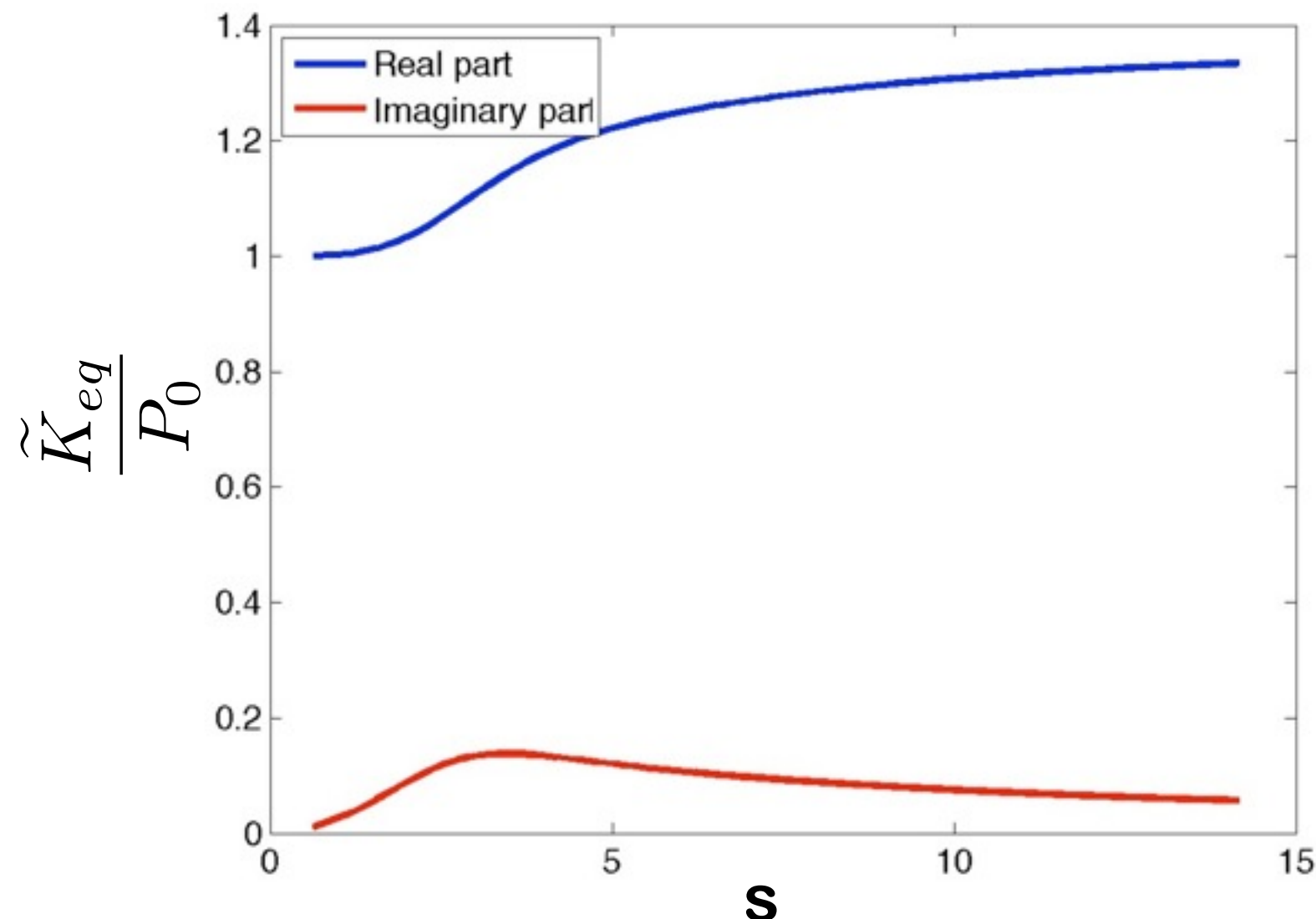
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$$\tilde{\beta}(\omega) = \frac{\gamma P_0}{\tilde{K}_{eq}(\omega)}$$

Dynamic compressibility

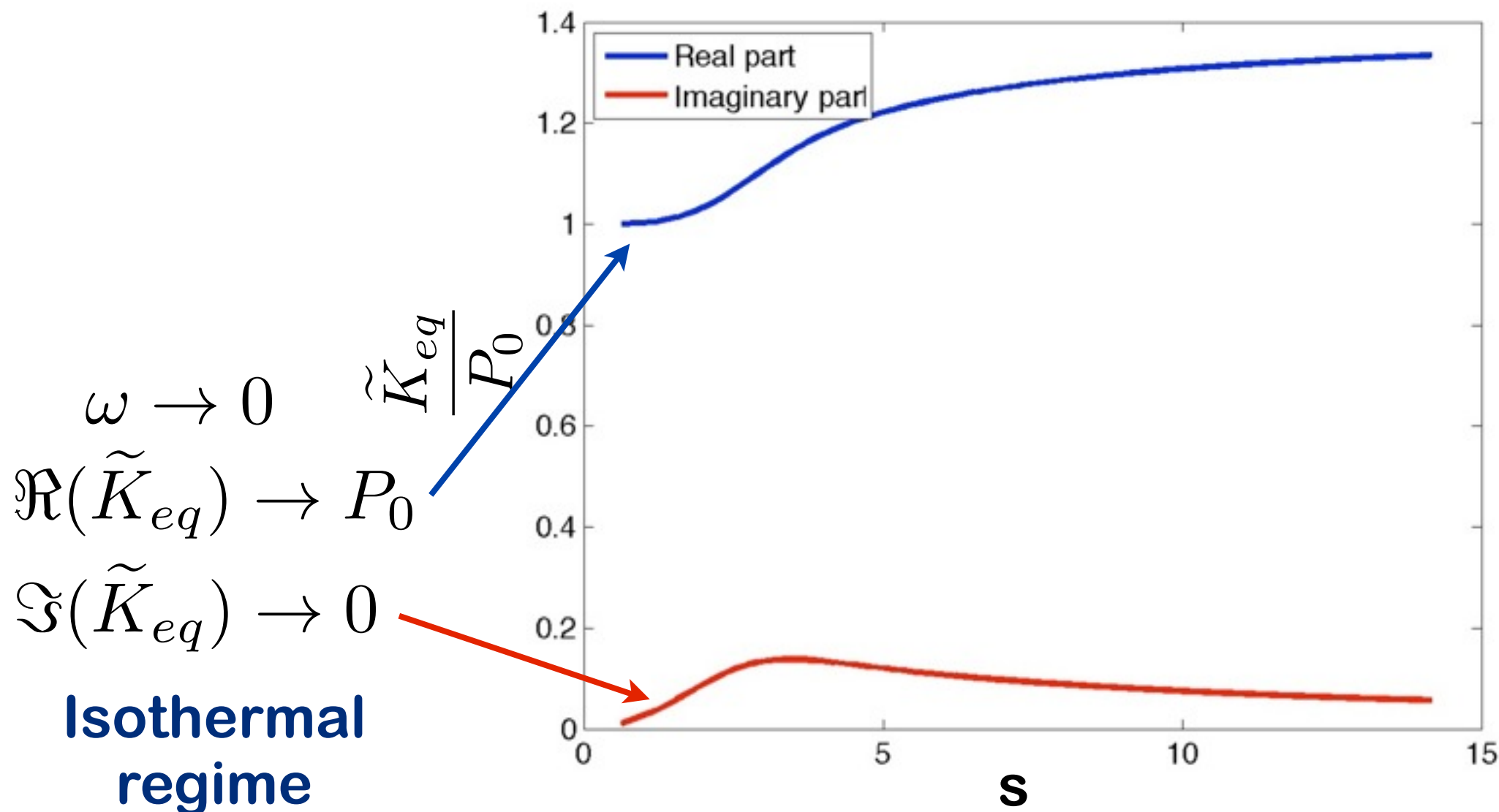


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Dynamic compressibility

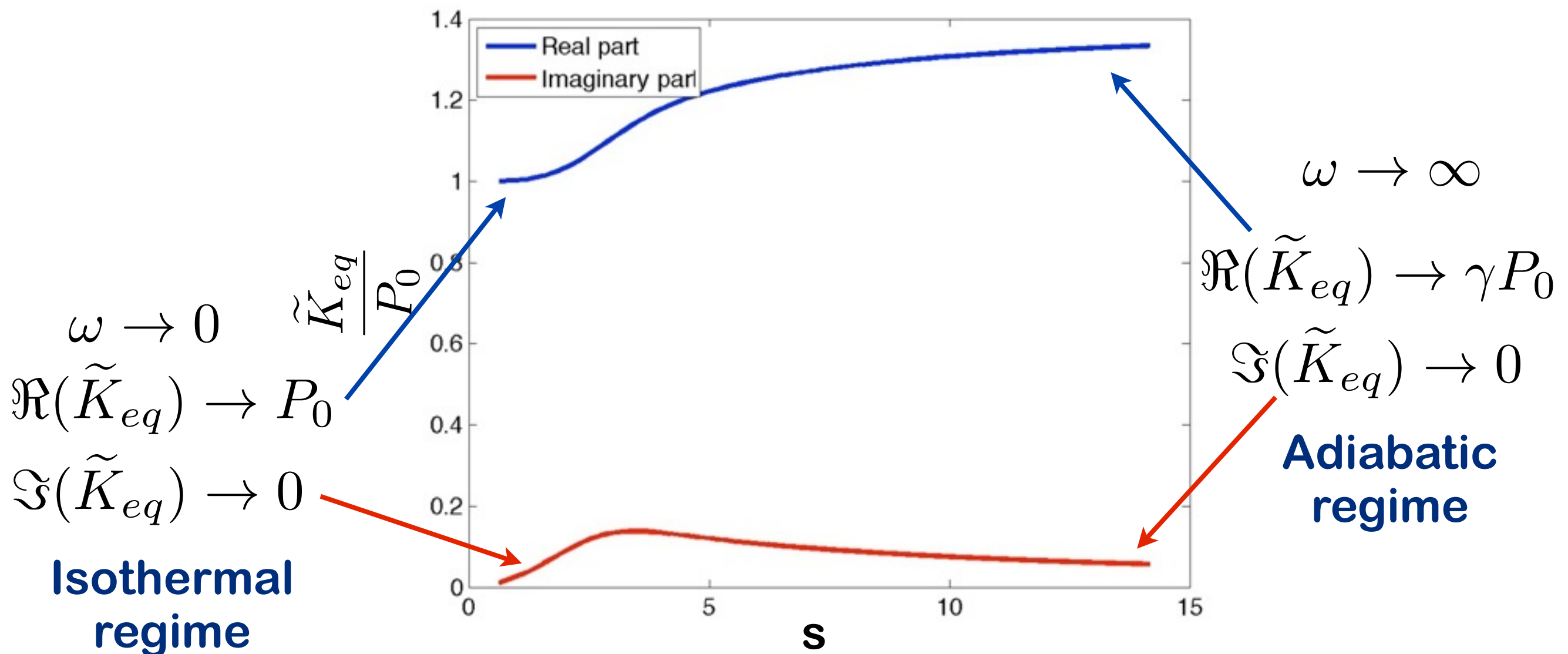


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Dynamic compressibility



$$\lim_{\omega \rightarrow 0} \tilde{K}_{eq}(\omega) = P_0 \left[1 + j\omega \frac{(\gamma - 1)Pr}{\gamma\nu} \frac{R^2}{8} \right] + O(\omega^2)$$

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**First order of real part
given by the saturating
fluid properties**

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$$k'_0 = \frac{R^2}{8} \quad \text{Thermal permeability}$$

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$$\lim_{\omega \rightarrow \infty} \tilde{K}_{eq}(\omega) = \gamma P_0 \left[1 + (-1 + j)(\gamma - 1) \frac{\delta_t}{R} \right] + O(\omega)$$

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$$\delta_t = \sqrt{\frac{2\nu}{Pr\omega}} = \sqrt{\frac{2\kappa}{\rho_0 C_P \omega}}$$

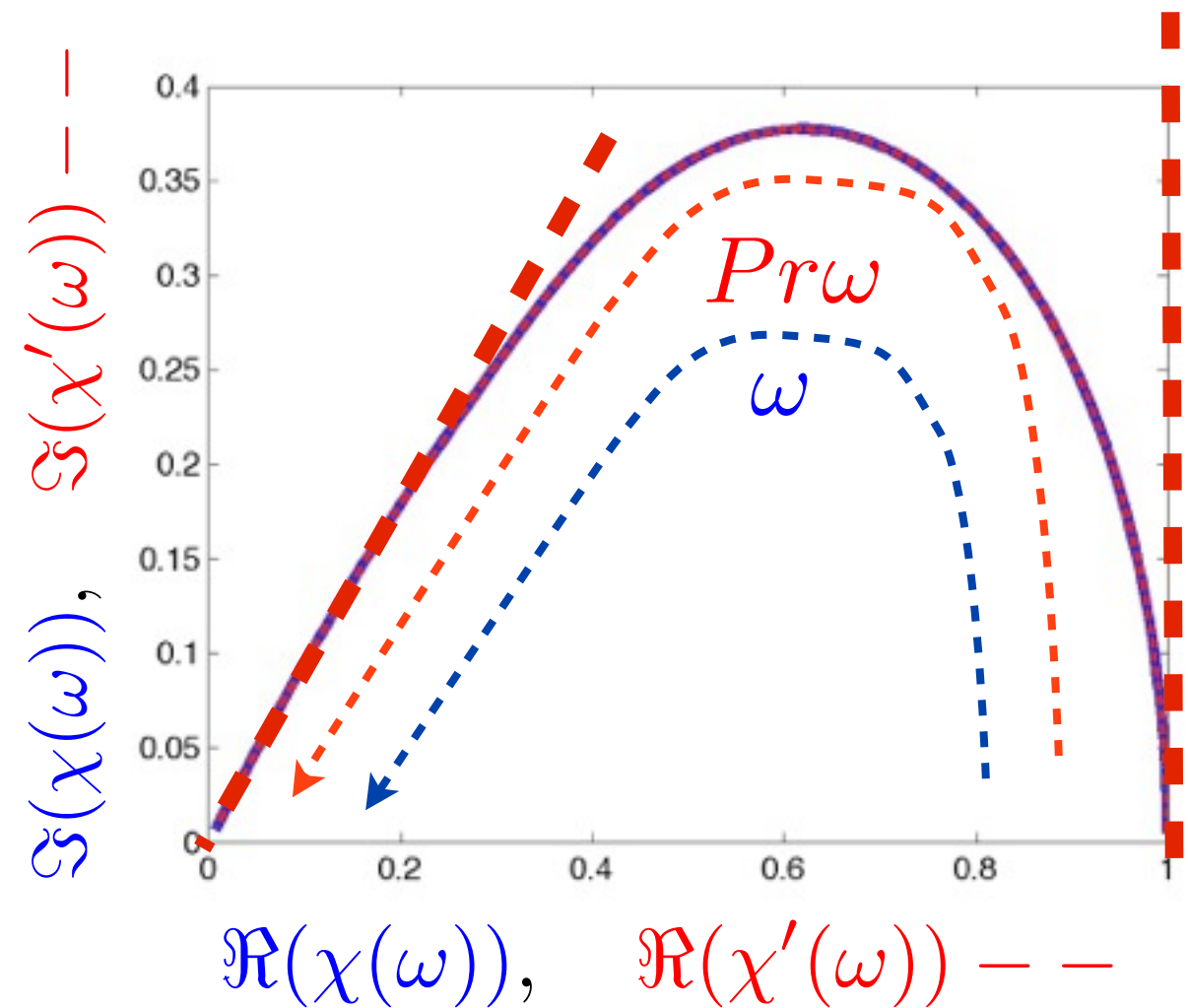
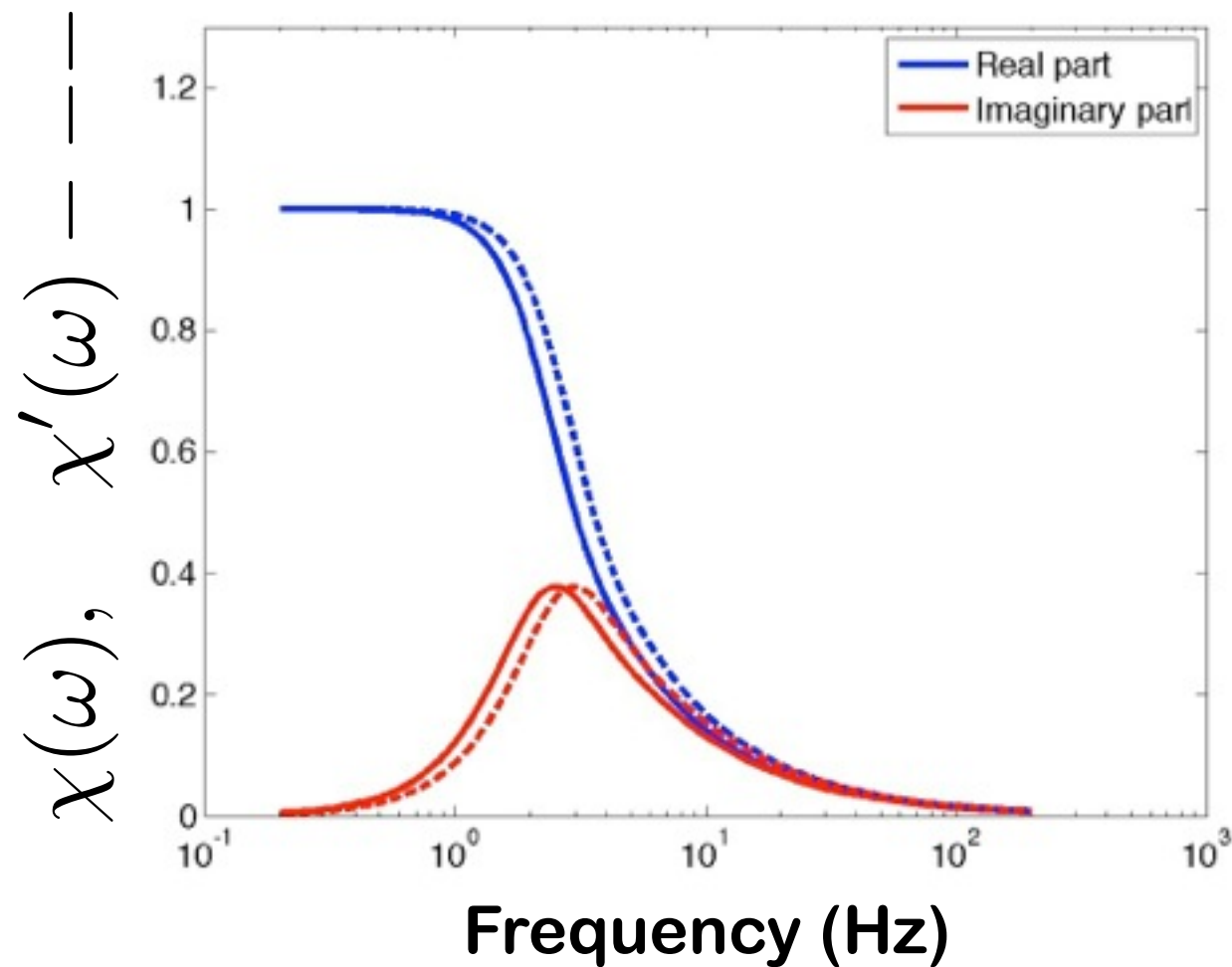
Thermal skin thickness

Concept of thermal
characteristic length

$$\Lambda' = R$$

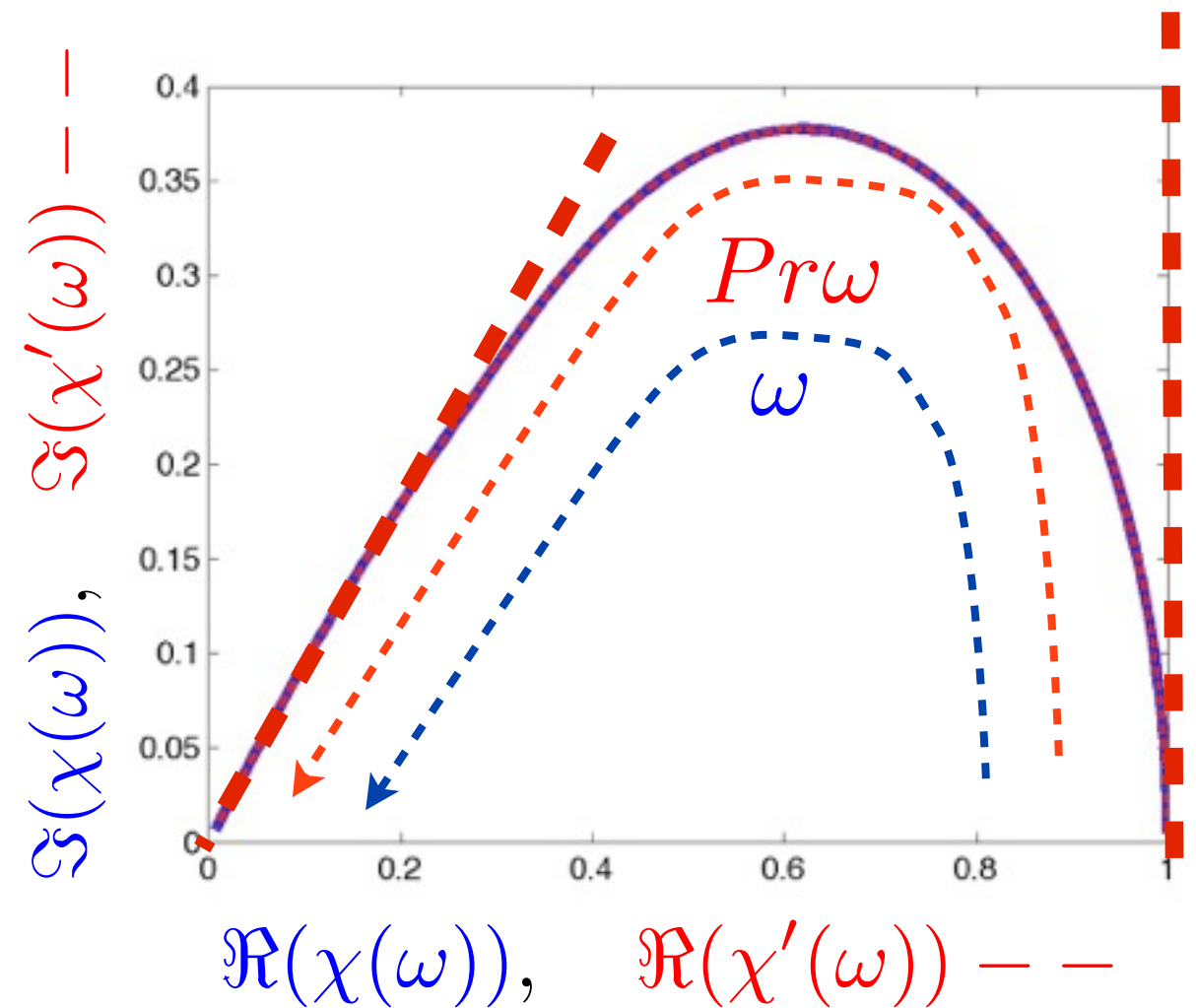
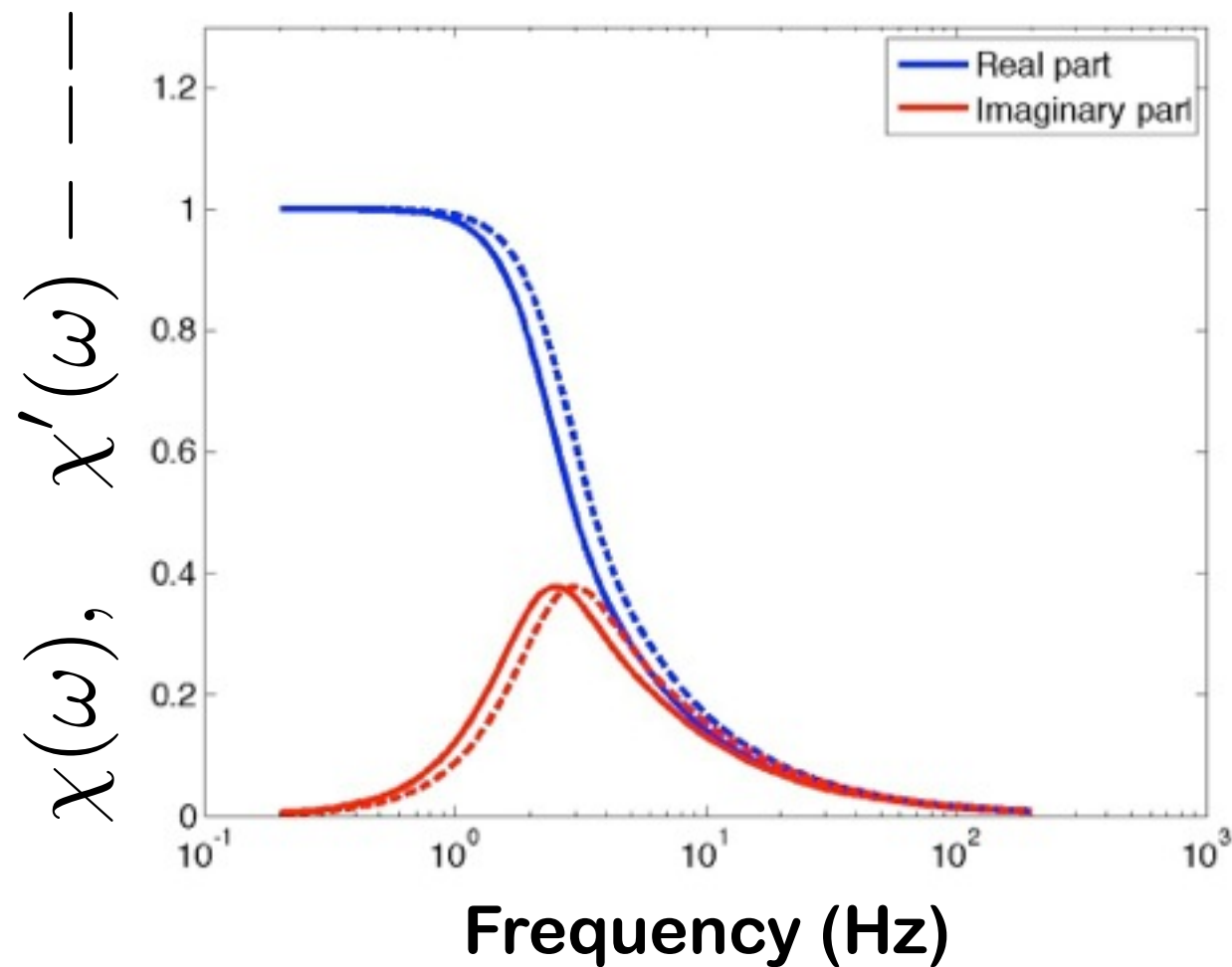
Thermal relaxation function

$$\chi'(\omega) = \frac{\tilde{\beta}(\omega) - 1}{\gamma - 1}$$

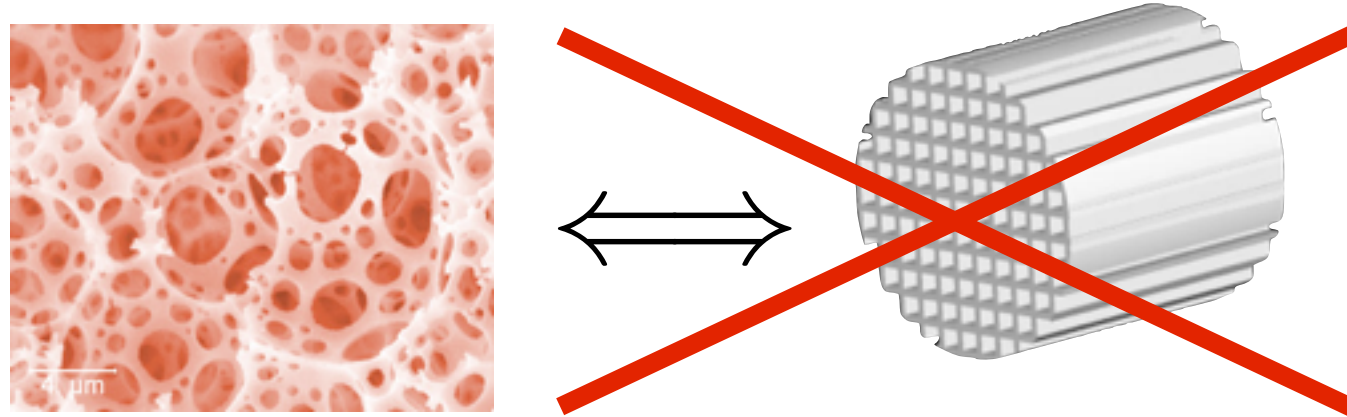


Thermal relaxation function

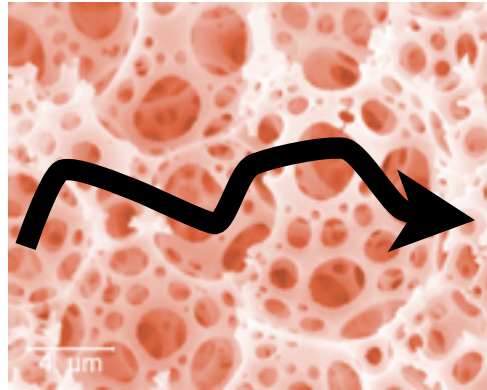
$$\chi'(\omega) = \frac{\tilde{\beta}(\omega) - 1}{\gamma - 1} = \chi \left(\frac{\omega}{Pr} \right)$$



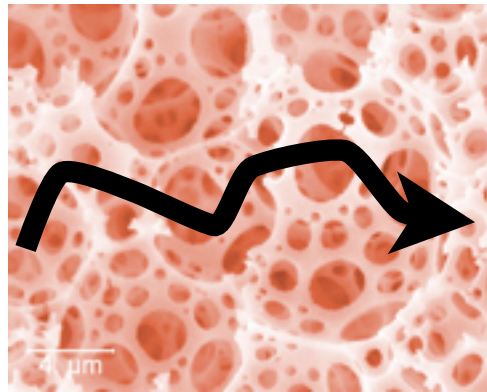
- **Analytical resolution for perfect geometries**
 - Decoupling of viscous and thermal effects
 - Homogenized quantities
- **LF and HF intrinsic parameters**
 - Exhibited in these simplified cases
 - Analysis on real and imaginary part of parameters
 - Some of them are more important than others
- **Relaxation function to link LF and HF**
 - Simple forms
 - Similarity of viscous and thermal relaxation



- **Real porous materials**
 - Complex and unknown skeleton geometry
 - No exact resolution of microscopic equations
- **Same physical mechanisms than previously**
 - Geometric models are particular cases
 - Setup by analogy
- **Intrinsic parameters**
 - Their number depends on desired accuracy
 - Relaxation functions

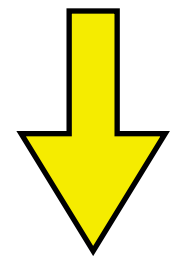


- Very high frequency parameter
- Even for perfect fluid
- Depends only on geometry



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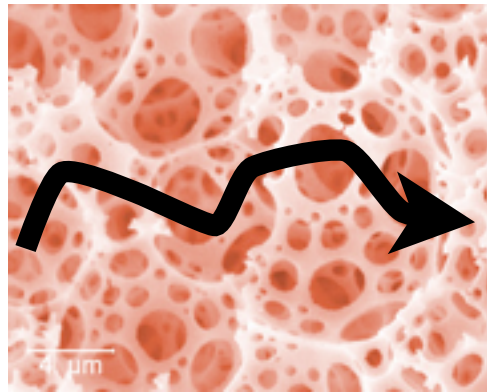
Microscopic kinetic energy on a REV Ω_f $\mathcal{T}_\mu = \frac{1}{2} \rho_0 v_\mu^2$



Averaging

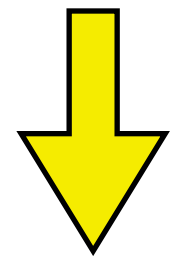
Macroscopic kinetic energy

$$\mathcal{T} = \langle \mathcal{T}_\mu \rangle_\Omega = \frac{1}{2} \rho_0 \phi \langle v_\mu^2 \rangle_{\Omega_f}$$



- Very high frequency parameter
- Even for perfect fluid
- Depends only on geometry

Microscopic kinetic energy on a REV Ω_f $\mathcal{T}_\mu = \frac{1}{2} \rho_0 v_\mu^2$



Averaging

Macroscopic kinetic energy

$$\begin{aligned} \mathcal{T} = \langle \mathcal{T}_\mu \rangle_\Omega &= \frac{1}{2} \rho_0 \phi \langle v_\mu^2 \rangle_{\Omega_f} = \frac{1}{2} \rho_0 \phi \alpha_\infty \langle v_\mu \rangle_{\Omega_f}^2 \\ &= \frac{1}{2} \frac{\rho_0 \alpha_\infty}{\phi} \|\vec{v}_{eq}\|^2 \end{aligned}$$

**Geometric tortuosity
dimensionless**

$$\alpha_\infty = \frac{\langle v_\mu^2 \rangle_{\Omega_f}}{\langle v_\mu \rangle_{\Omega_f}^2}$$

First approach: case of density

$\tilde{\rho}_{eq}$	Real part		Imaginary part	
	Parameter	Order	Parameter	Order
LF			k_0, σ	$(j\omega)^{-1}$
HF	a_∞	$(j\omega)^0$		

First approach: case of density

$\tilde{\rho}_{eq}$	Real part		Imaginary part	
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$$\tilde{\rho}_{eq}(\omega) = \frac{\rho_0}{\phi} \tilde{\alpha}(\omega) = \frac{\eta}{j\omega \tilde{k}(\omega)}$$

First approach: case of density

$\tilde{\rho}_{eq}$	Real part		Imaginary part	
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$$\lim_{\omega \rightarrow 0} \tilde{k}(\omega) = k_0 + O(j\omega) \quad \Bigg| \quad \lim_{\omega \rightarrow \infty} \tilde{\alpha}(\omega) = \alpha_\infty + O(j\omega)^{-1/2}$$

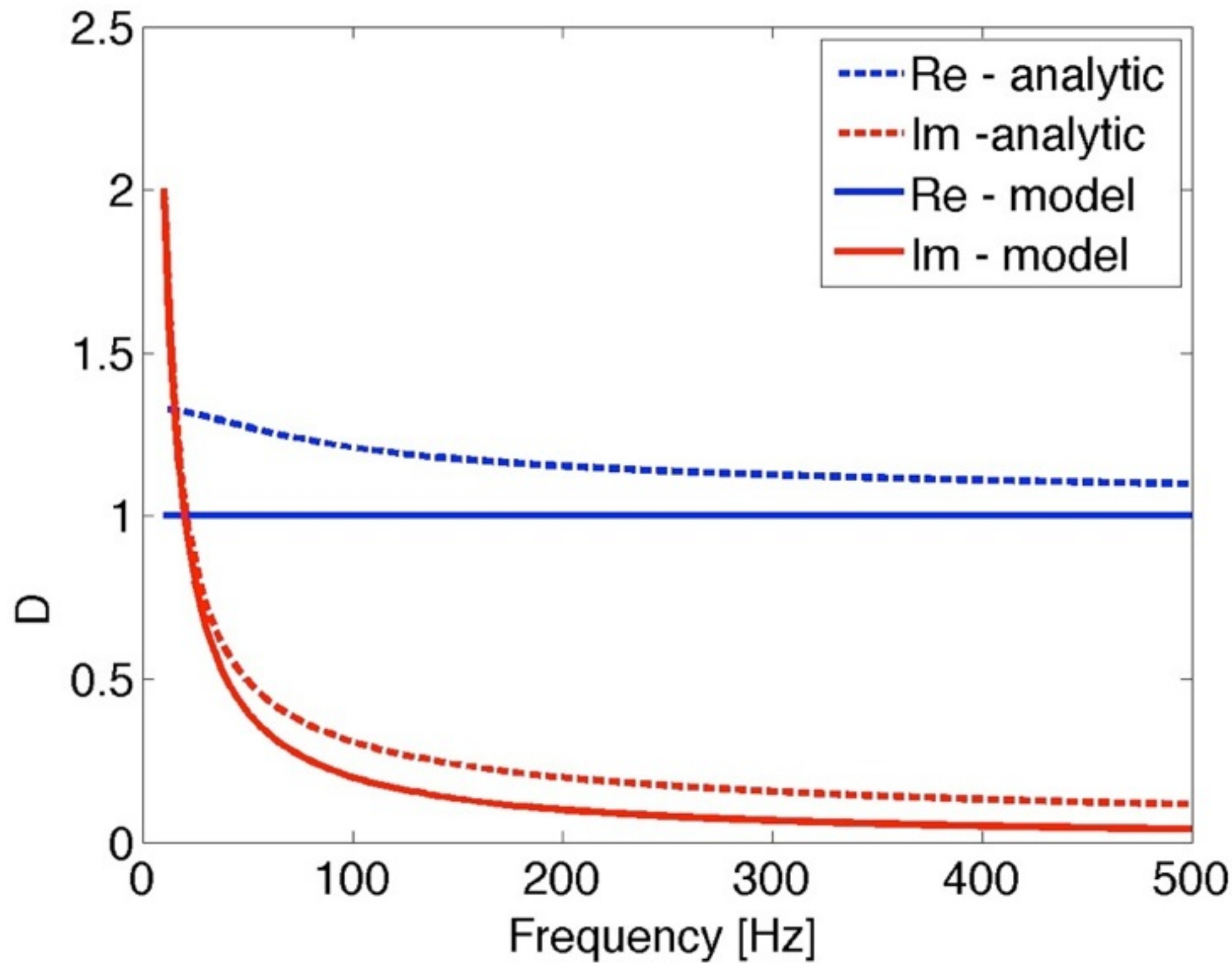
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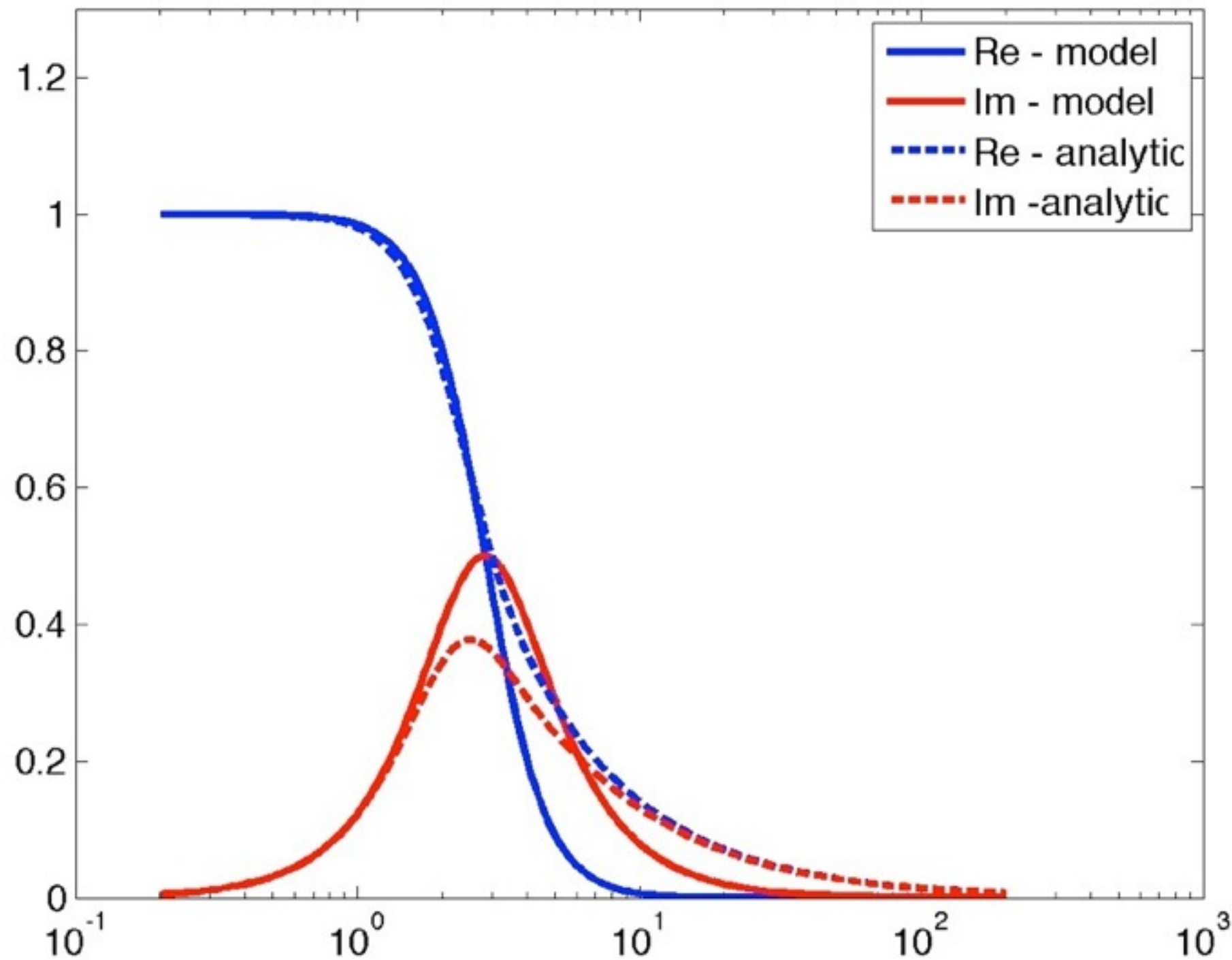
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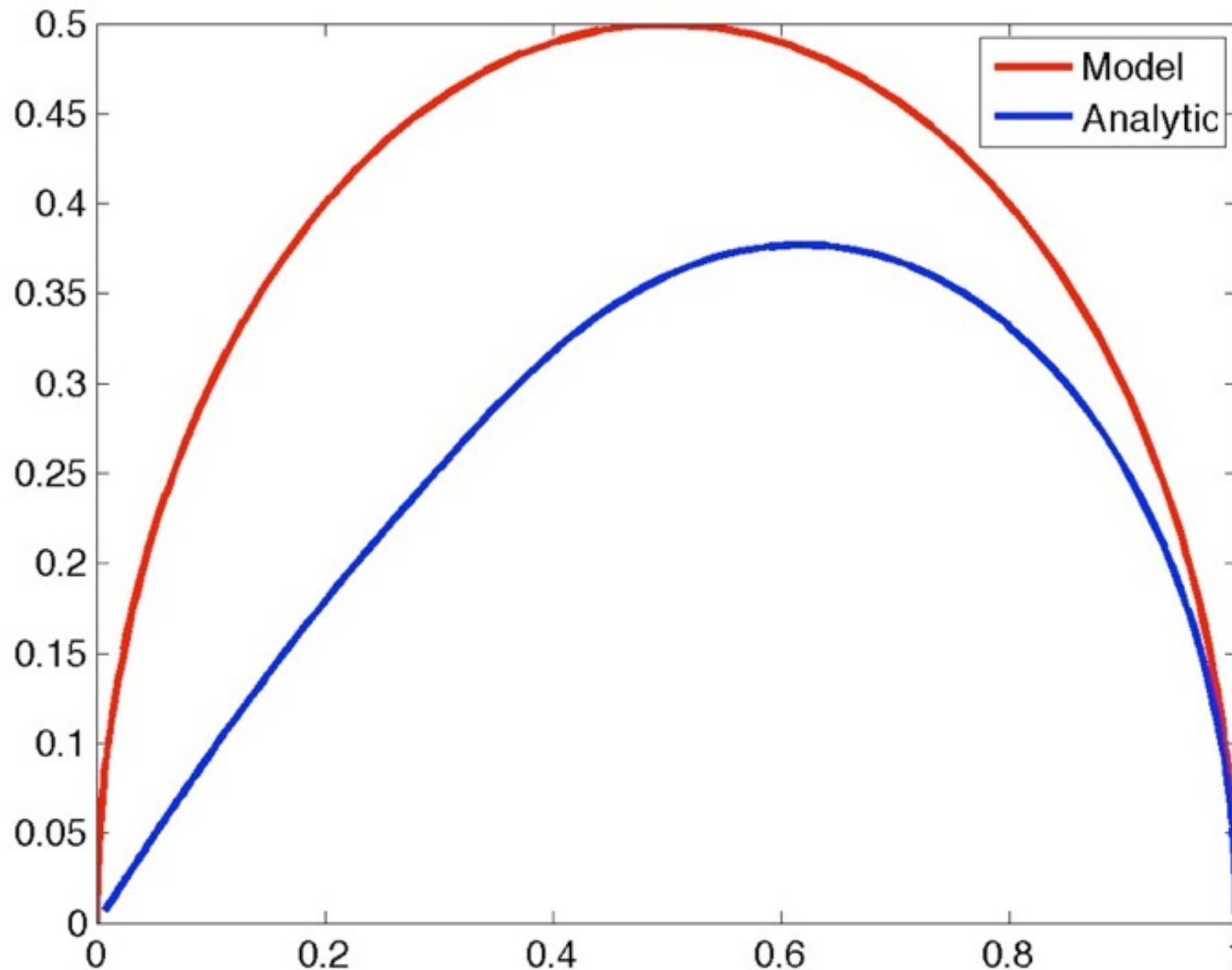
$$\lim_{\omega \rightarrow 0} \tilde{k}(\omega) = k_0 + O(j\omega) \quad \left| \quad \lim_{\omega \rightarrow \infty} \tilde{\alpha}(\omega) = \alpha_\infty + O(j\omega)^{-1/2}\right.$$

Let's try

$$\tilde{\alpha}(\omega) = \alpha_\infty \left[1 + \frac{\omega_0}{j\omega} \right] \quad \text{with} \quad \omega_0 = \frac{\sigma \phi}{\rho_0 \alpha_\infty}$$







Refinement needed at high frequencies

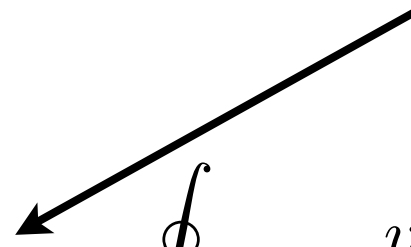
- High frequency parameters
- Viscous Characteristic Length Λ
- Thermal Characteristic Length Λ'

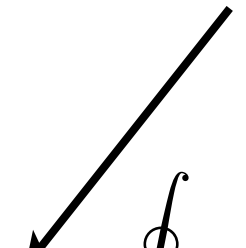
- High frequency parameters
- Viscous Characteristic Length Λ
- Thermal Characteristic Length Λ'

$$\lim_{\omega \rightarrow \infty} \chi(\omega) = (1 - j) \frac{\delta_v}{\Lambda} \quad \left| \quad \lim_{\omega \rightarrow \infty} \chi'(\omega) = (1 - j) \frac{\delta_t}{\Lambda'} \right.$$

- High frequency parameters
- Viscous Characteristic Length Λ
- Thermal Characteristic Length Λ'

$$\lim_{\omega \rightarrow \infty} \chi(\omega) = (1 - j) \frac{\delta_v}{\Lambda} \quad \Bigg| \quad \lim_{\omega \rightarrow \infty} \chi'(\omega) = (1 - j) \frac{\delta_t}{\Lambda'}$$


$$\frac{2}{\Lambda} = \frac{\oint_{\partial\Omega_{pores}} v^2(\mathbf{r}_p) dA}{\int_{\Omega_{pores}} v^2(\mathbf{r}) d\Omega}$$


$$\frac{2}{\Lambda'} = \frac{\oint_{\partial\Omega_{pores}} dA}{\int_{\Omega_{pores}} d\Omega}$$

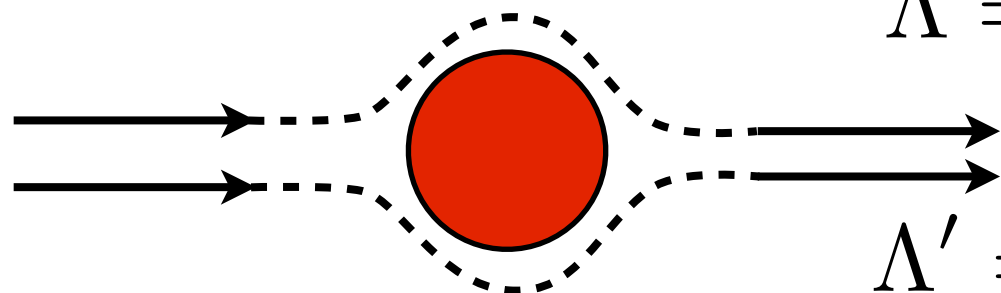
- High frequency parameters
- Viscous Characteristic Length Λ
- Thermal Characteristic Length Λ'

$$\lim_{\omega \rightarrow \infty} \chi(\omega) = (1 - j) \frac{\delta_v}{\Lambda} \quad \Bigg| \quad \lim_{\omega \rightarrow \infty} \chi'(\omega) = (1 - j) \frac{\delta_t}{\Lambda'}$$

$$\frac{2}{\Lambda} = \frac{\oint_{\partial\Omega_{pores}} v^2(\mathbf{r}_p) dA}{\int_{\Omega_{pores}} v^2(\mathbf{r}) d\Omega}$$

$$\frac{2}{\Lambda'} = \frac{\oint_{\partial\Omega_{pores}} dA}{\int_{\Omega_{pores}} d\Omega}$$

Example



$$\Lambda = \frac{1}{2\pi LR}$$

$$\Lambda' = \frac{1}{\pi LR} = 2\Lambda$$

In general

$$\Lambda' > \Lambda$$

$\tilde{\rho}_{eq}$	Real part		Imaginary part	
	Parameter	Order	Parameter	Order
LF			k_0, σ	$(j\omega)^{-1}$
HF	α_∞	$(j\omega)^0$	Λ	$(j\omega)^{-1/2}$
$\tilde{K}_{eq}$	Real part		Imaginary part	
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Property of
saturating fluid

$\tilde{\rho}_{eq}$	Real part		Imaginary part	
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LF			k_0, σ	$(j\omega)^{-1}$
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Property of
saturating fluid

Order $1/2$ terms
(dissipation)

$$\tilde{\alpha}(\omega) = \alpha_{\infty} \left[1 + \frac{\omega_0}{j\omega} F(\omega) \right]$$

$$\tilde{\alpha}(\omega) = \alpha_{\infty} \left[1 + \frac{\omega_0}{j\omega} \underbrace{F(\omega)}_{\text{Shape function}} \right]$$

$$\tilde{\alpha}(\omega) = \alpha_{\infty} \left[1 + \frac{\omega_0}{j\omega} \overset{\text{Shape function}}{F(\omega)} \right]$$

Second term in the HF tortuosity expansion

$$\lim_{\omega \rightarrow \infty} \tilde{\alpha}(\omega) = \alpha_{\infty} \left[1 + \frac{1}{\Lambda} \sqrt{\frac{4\nu}{j\omega}} \right]$$

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LF and HF limits of the shape function

$$F(0) = 1 \qquad \lim_{\omega \rightarrow \infty} F(\omega) = \sqrt{\frac{j\omega}{\omega_{\infty}}} \qquad \text{with } \omega_{\infty} = \frac{\sigma^2 \phi^2 \Lambda^2}{4\eta \rho_0 \alpha_{\infty}^2}$$

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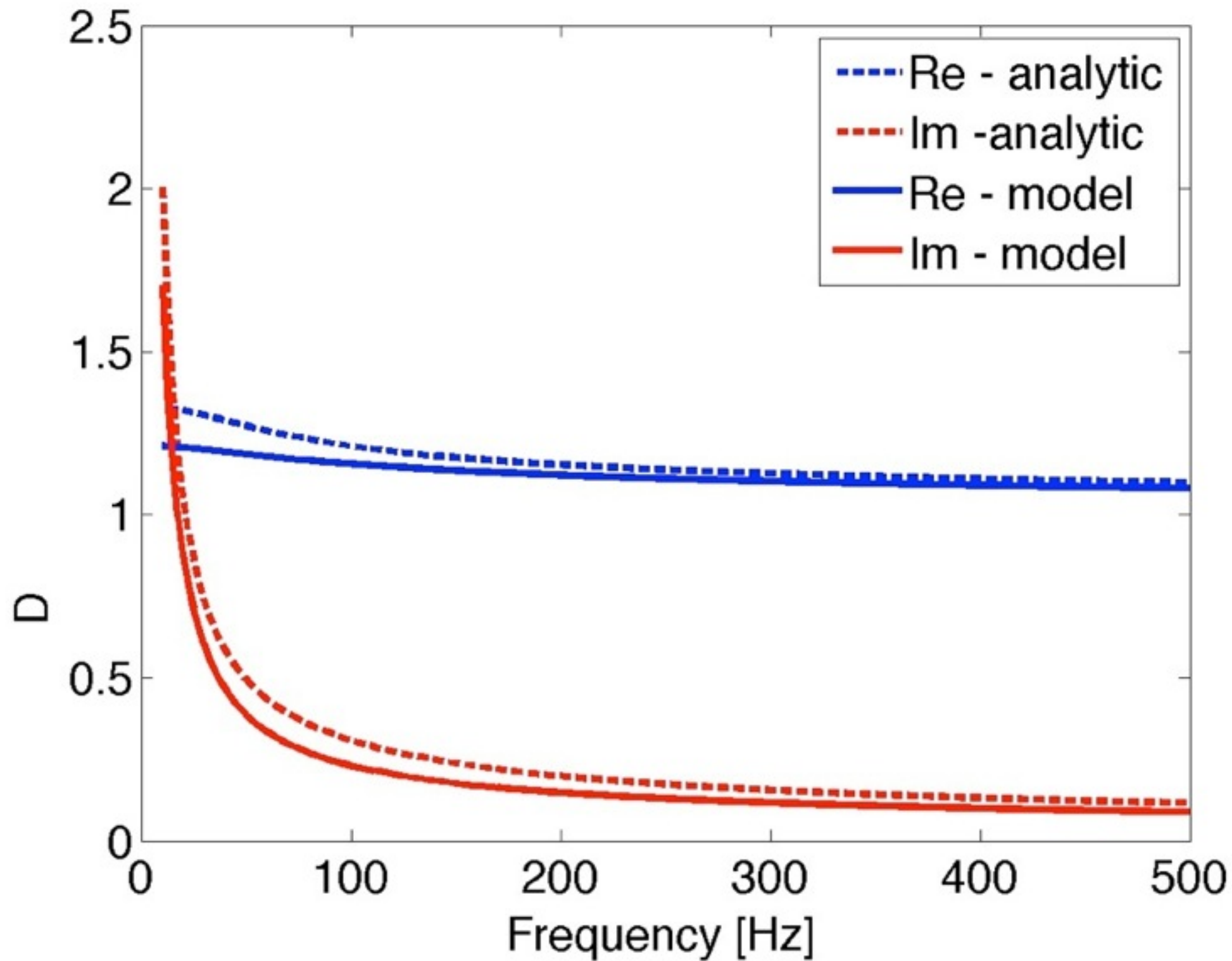
Second term in the HF tortuosity expansion

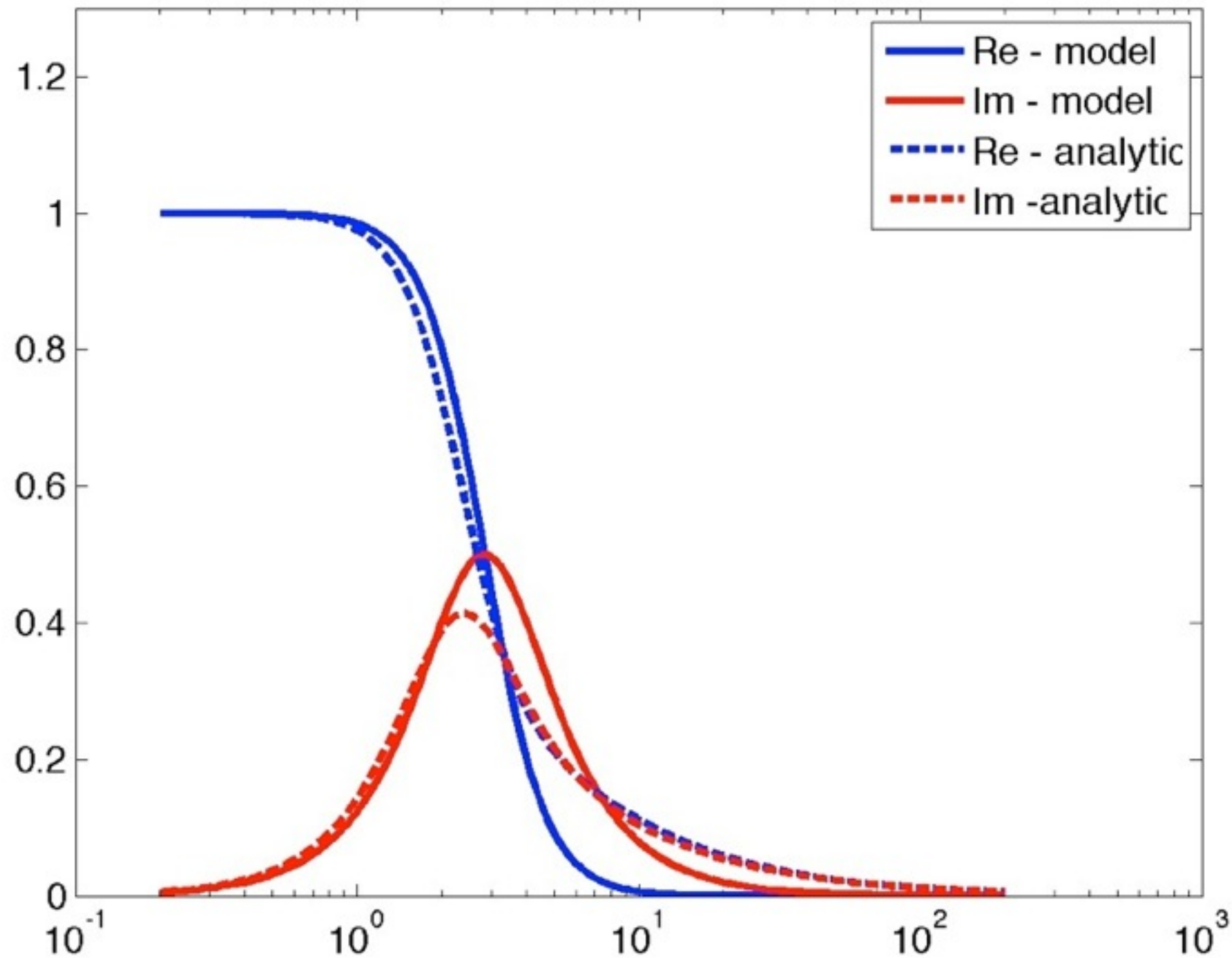
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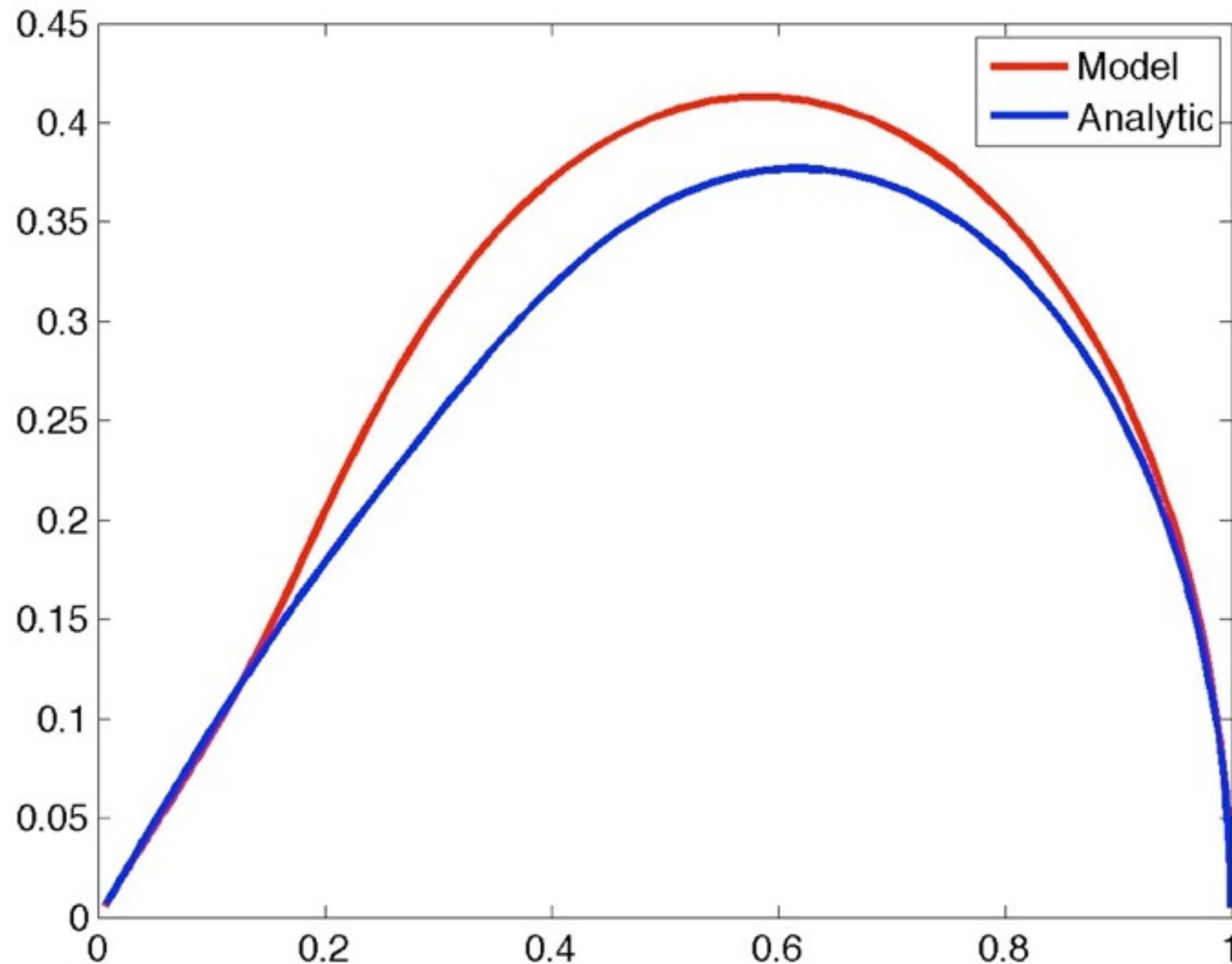
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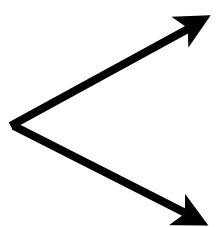
$$F(\omega) = \sqrt{1 + \frac{j\omega}{\omega_{\infty}}} \quad \tilde{\rho}_{eq}(\omega) = \frac{\rho_0 \alpha_{\infty}}{\phi} \left[1 + \frac{\omega_0}{j\omega} \sqrt{1 + \frac{j\omega}{\omega_{\infty}}} \right]$$







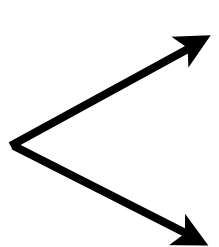
Quite Ok but error on the LF real part

A first comment  **Density: 3 parameters σ , α_∞ , Λ**
Compressibility: 1 parameter Λ'

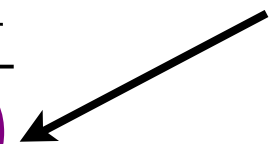
A first comment $\rightarrow$ **Density: 3 parameters** σ , α_∞ , Λ
 $\rightarrow$ **Compressibility: 1 parameter** Λ'

$$\tilde{K}_{eq}(\omega) = \frac{\gamma P_0 / \phi}{\gamma - \frac{\gamma - 1}{\alpha'(\omega)}}$$

Thermal tortuosity $\rightarrow$ $\alpha'(\omega)$

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$$\tilde{K}_{eq}(\omega) = \frac{\gamma P_0 / \phi}{\gamma - \frac{\gamma - 1}{\alpha'(\omega)}}$$

Thermal tortuosity 

$$\alpha'(\omega) = 1 + \frac{\omega'_\infty}{2j\omega} F'(\omega) \quad \text{with} \quad \omega'_\infty = \frac{16\nu'}{\Lambda'^2}$$

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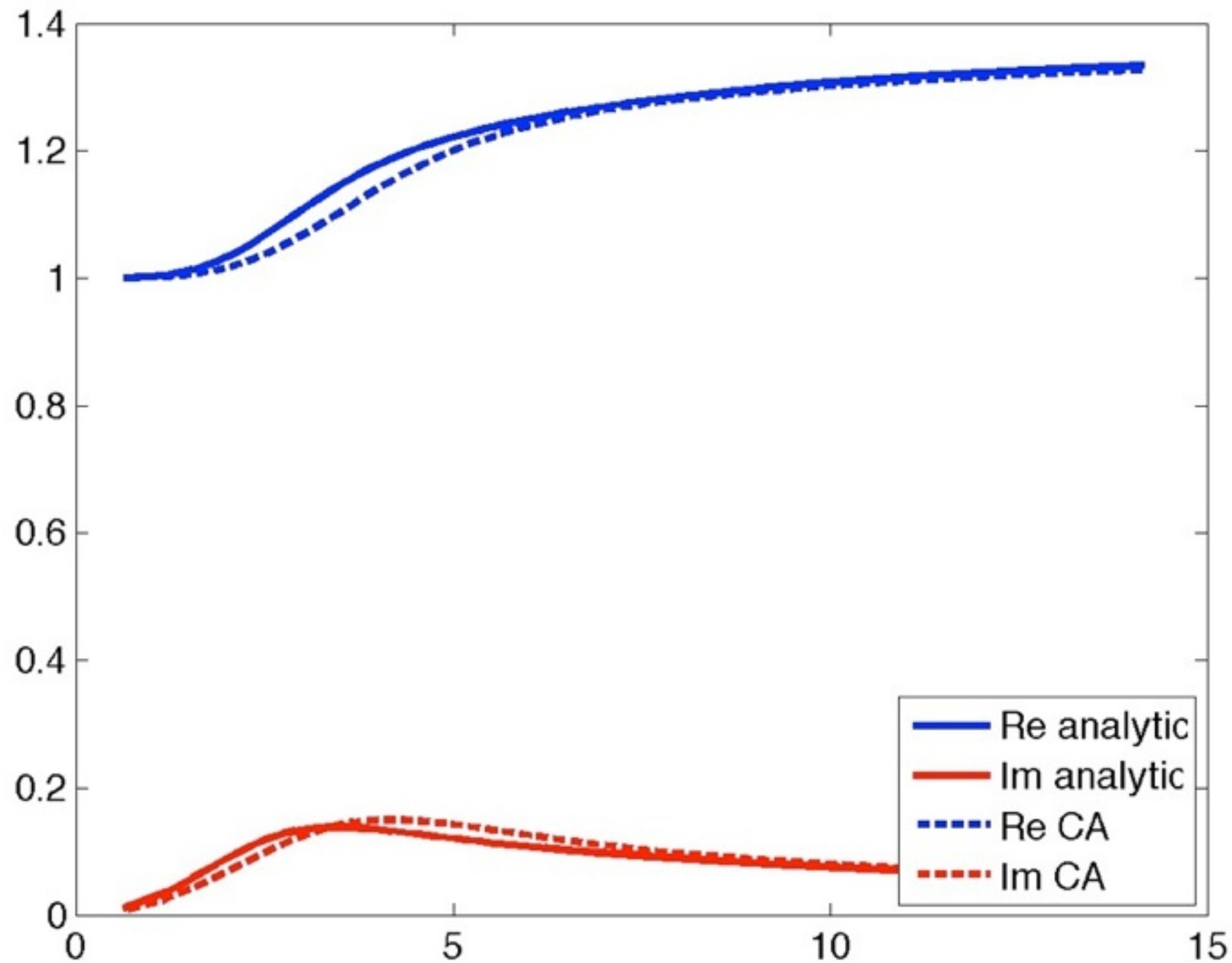
Thermal tortuosity $\rightarrow$

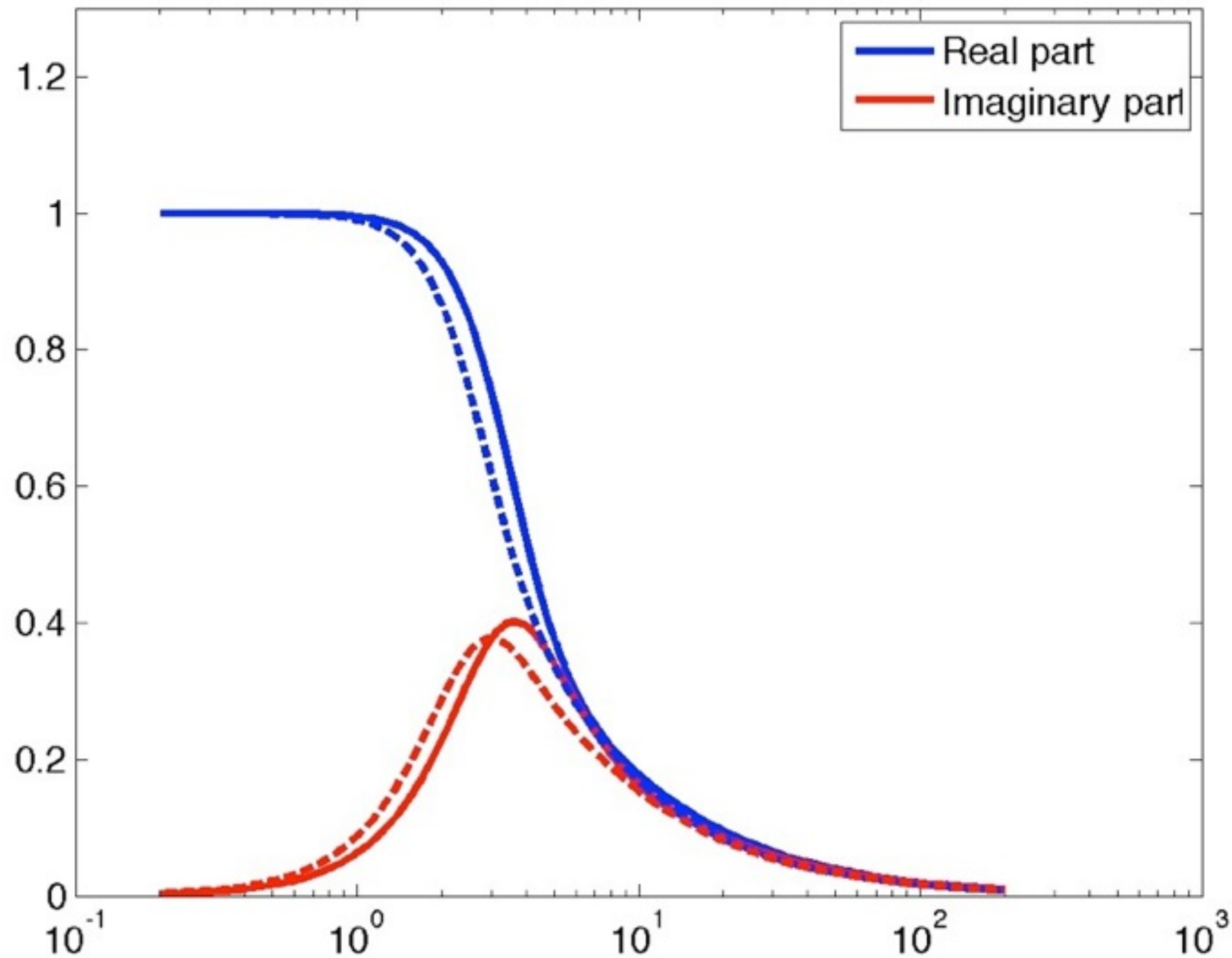
$$\alpha'(\omega) = 1 + \frac{\omega'_\infty}{2j\omega} F'(\omega) \quad \text{with} \quad \omega'_\infty = \frac{16\nu'}{\Lambda'^2}$$

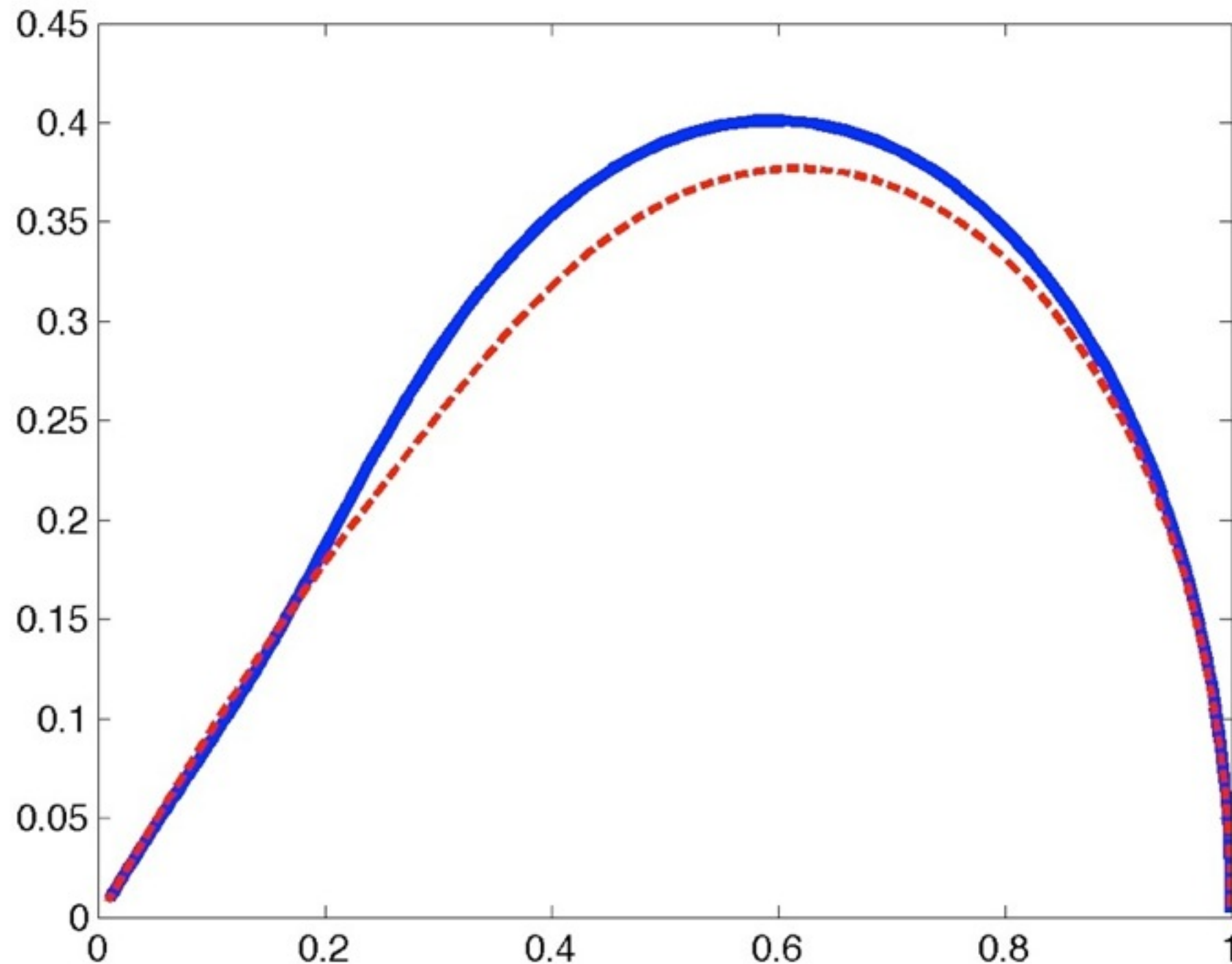
Possible choice:

$$F'(\omega) = \sqrt{1 + \frac{j\omega}{\omega'_\infty}}$$

$$\tilde{K}_{eq}(\omega) = \frac{\gamma P_0 / \phi}{\gamma - \frac{\gamma - 1}{1 + \frac{\omega'_\infty}{2j\omega} \sqrt{1 + \frac{j\omega}{\omega'_\infty}}}}$$







Sufficient for practical use

$$\tilde{\alpha}(\omega) = \alpha_{\infty} \left[1 + \frac{\omega_0}{j\omega} F(\omega) \right]$$

$$\lim_{\omega \rightarrow \infty} F(\omega) = \sqrt{\frac{j\omega}{\omega_{\infty}}}$$

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Refinement of the model at low frequency

$$\lim_{\omega \rightarrow 0} F(\omega) = 1 + \frac{j\omega}{\omega_0} \beta$$

← Measure of effective fluid-inertia enhancement at LF

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$$F(\omega) = 1 - p + p \sqrt{1 + \frac{j\omega}{\omega_{\infty} p^2}} \quad \text{with} \quad p = \frac{\omega_0}{2\beta\omega_{\infty}}$$

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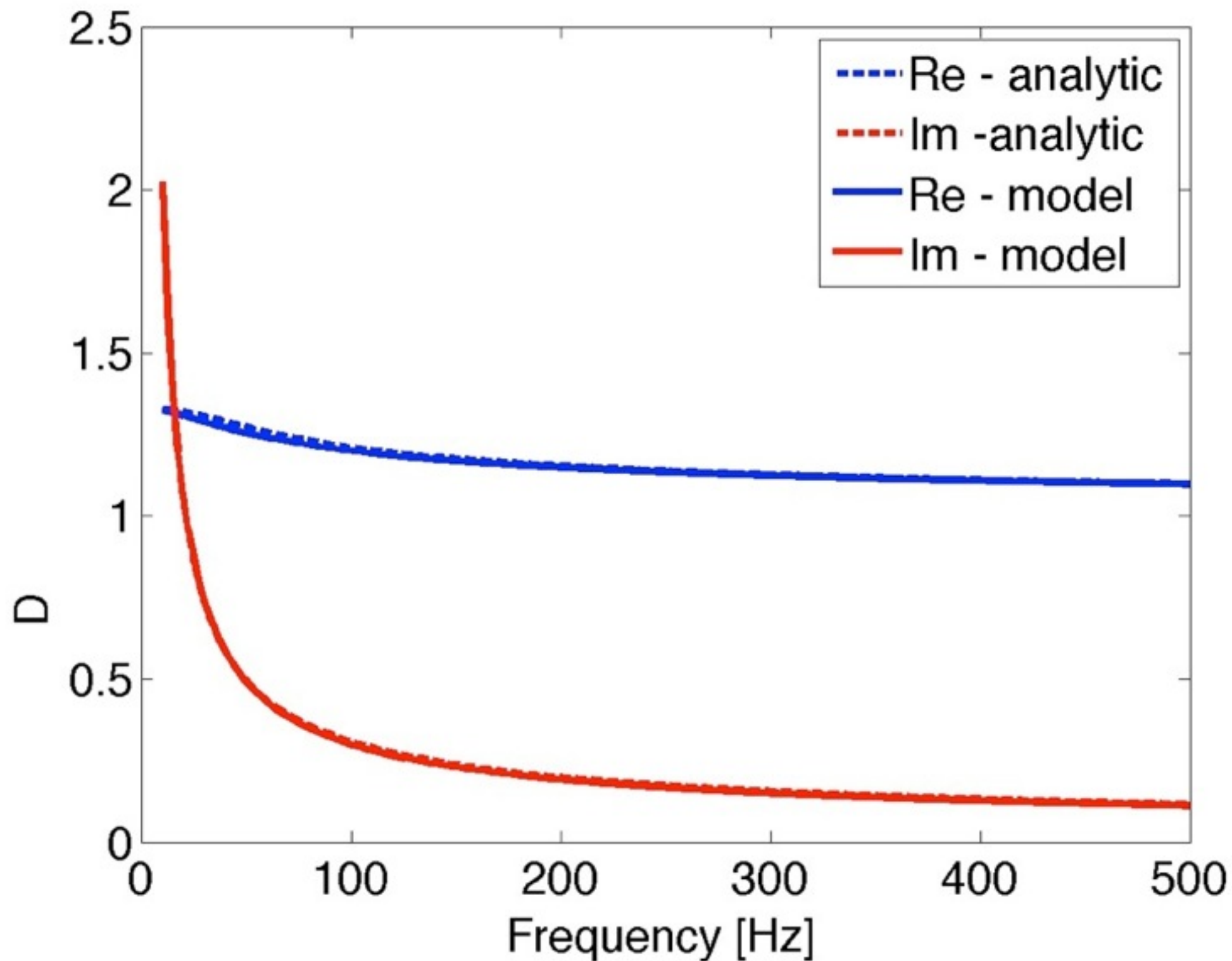
Refinement of the model at low frequency

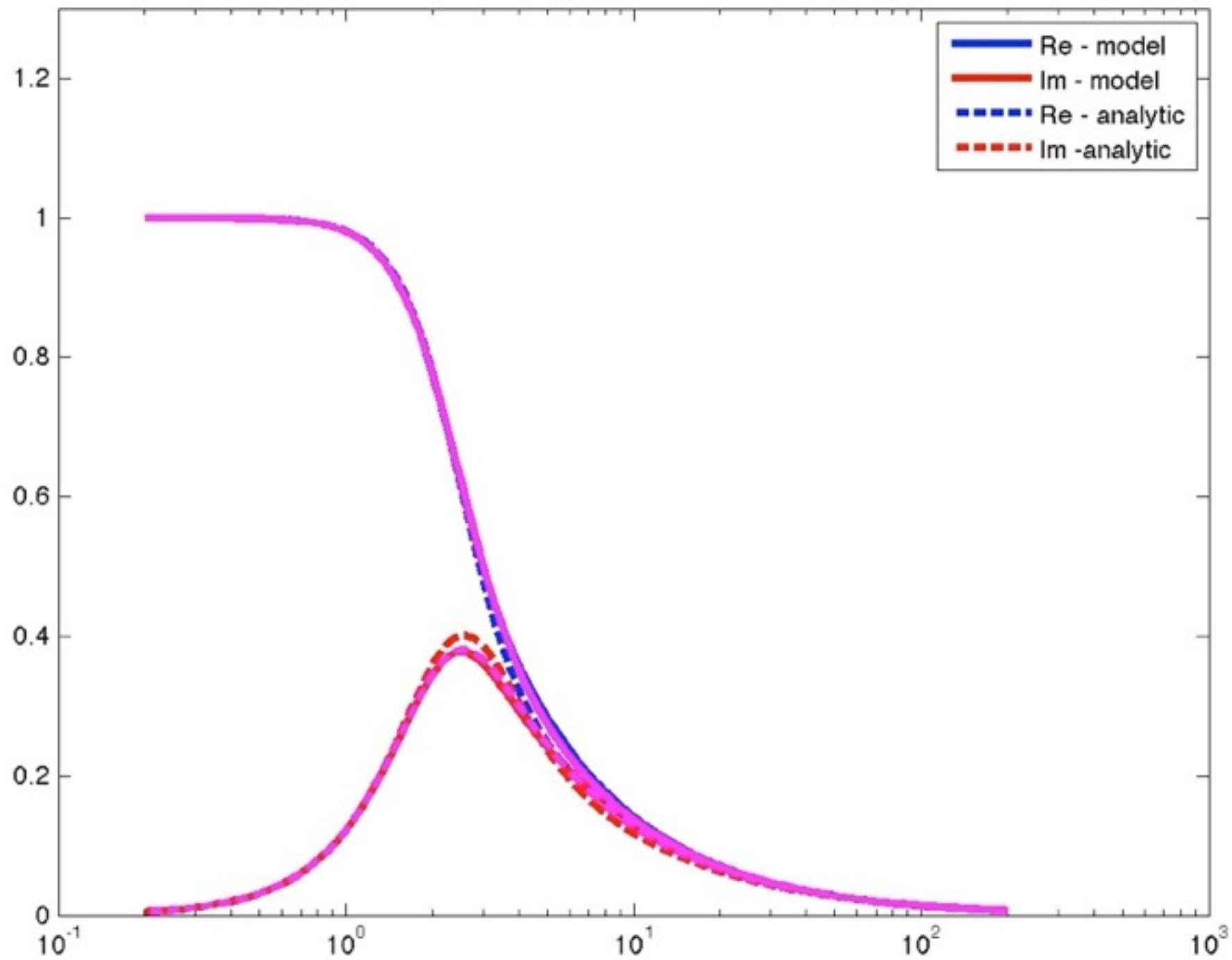
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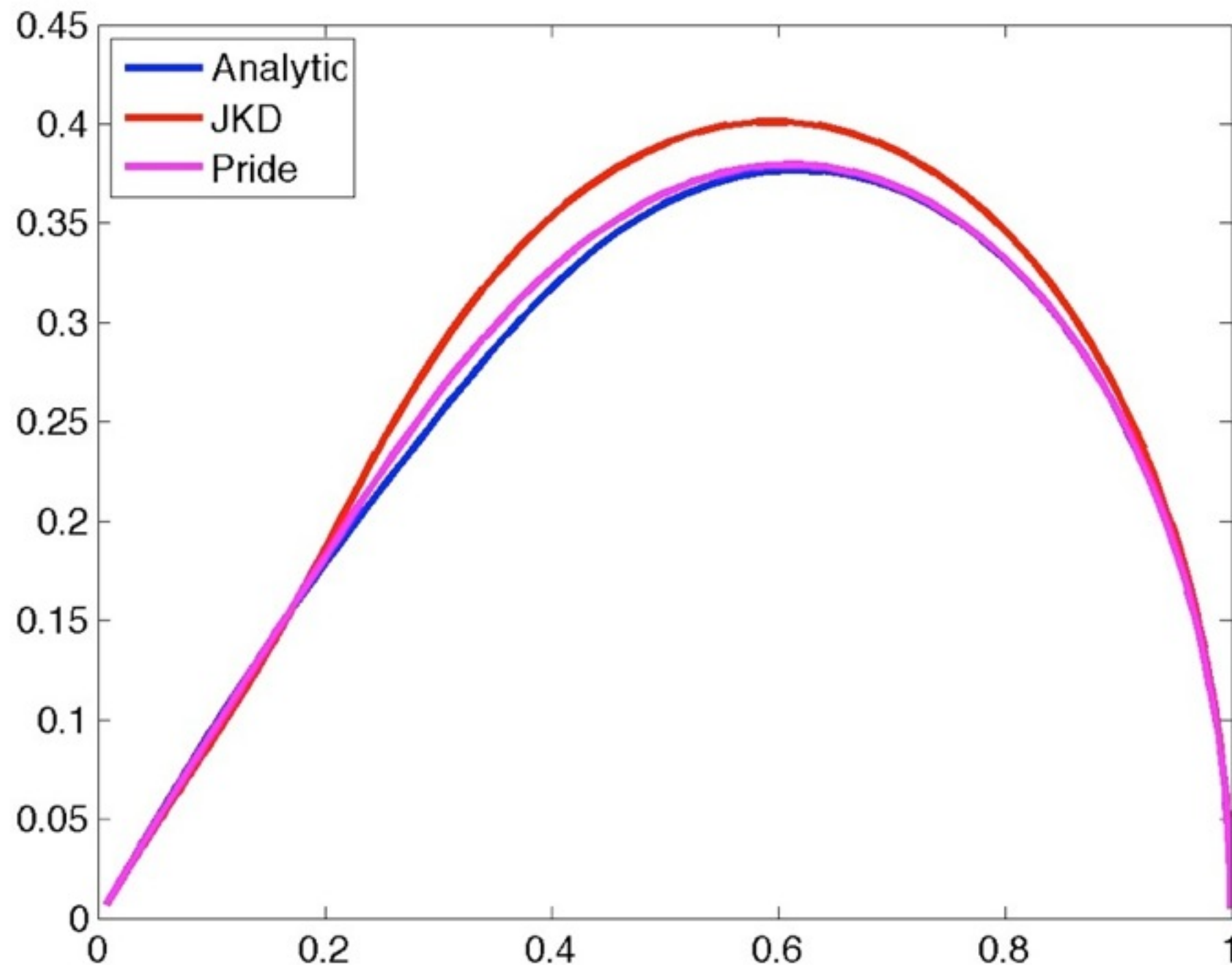
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$$\tilde{\rho}_{eq}(\omega) = \frac{\rho_0 \alpha_{\infty}}{\phi} \left[1 + \frac{\omega_0}{j\omega} \left(1 - p + p \sqrt{1 + \frac{j\omega}{\omega_{\infty} p^2}} \right) \right]$$





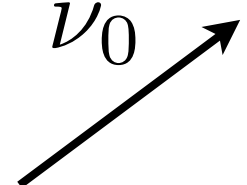


Nearly exact

LF second order term (imaginary part)

$$\lim_{\omega \rightarrow 0} \tilde{K}_{eq}(\omega) = \frac{P_0}{\phi} \left[1 + j\omega \frac{\gamma - 1}{\gamma} \frac{Pr \rho_0}{\nu_0} k'_0 \right]$$

Thermal permeability



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Separation of LF and HF param. Thermal permeability

$$\alpha'(\omega) = 1 + \frac{\omega'_0}{2j\omega} \sqrt{1 + \frac{j\omega\omega'_\infty}{\omega'^2_0}} \quad \omega'_0 = \frac{2\phi\nu'}{k'_0}$$

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$$\tilde{K}_{eq}(\omega) = \frac{\gamma P_0 / \phi}{\gamma - \frac{1}{1 + \frac{\omega'_0}{2j\omega} \sqrt{1 + \frac{j\omega\omega'_\infty}{\omega'^2_0}}}}$$

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One more thing: Refined Lafarge model

Lafarge PhD (1995)

$$\lim_{\omega \rightarrow 0} \tilde{K}_{eq}(\omega) = P_0 \left[1 + j\omega \frac{\gamma - 1}{\gamma} \frac{Pr \rho_0}{\nu_0} k'_0 - \omega^2 \frac{\gamma - 1}{\gamma} \frac{k'^2_0 Pr^2 \rho_0^2}{\nu_0^2} \alpha'_0 \right]$$

Thermal tortuosity

$\tilde{\rho}_{eq}$	Real part		Imaginary part	
	Parameter	Order	Parameter	Order
LF	α_0, p	$(j\omega)^0$	k_0, σ	$(j\omega)^{-1}$
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$\tilde{K}_{eq}$	Real part		Imaginary part	
	Parameter	Order	Parameter	Order
LF	P_0	$(j\omega)^0$	k'_0	$(j\omega)^1$
	α'_0	$(j\omega)^2$		
HF	γP_0	$(j\omega)^0$	Λ'	$(j\omega)^{-1/2}$

- **An equivalent fluid is a “classical” Helmholtz fluid**
 - Dissipation
 - Dispersion
 - Frequency dependence
- **Decoupling of dissipative effects**
 - Viscosity: density
 - Thermal effects: compressibility
- **Several kinds of models**
 - Empirical
 - Geometrical
 - Semi-phenomenological